

Tracking country innovation performance: The Innovation Output Indicator 2025

Bello, M., Ravanos, P., Tessarin, M.

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Contact information

Name: Michela Bello

Address: Via E. Fermi 2749, TP 361, Ispra (VA), I-21027, Italy

Email: Michela.BELLO@ec.europa.eu

Tel.: 0039 033278 3976

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Abstract

This report presents the latest edition of the Innovation Output Indicator (IOI) for 2025, a composite indicator published by the European Commission since 2013 to offer an output-focused metric of innovation performance at both national and European Union (EU) levels. The IOI measures countries' capacity to derive economic benefits from innovation by tracking the extent to which innovative ideas reach the market, create knowledge-intensive jobs, and increase country technological capability. It was developed in response to a 2011 European Council request to enhance the monitoring of innovation progress. It also serves as a flagship indicator for assessing the progress of the underlying objectives of the EU Framework Programmes for Research and Innovation and contributes to the European Commission's Directorate-General for Research and Innovation (DG RTD) assessment of innovation performance in the EU, complementing the European Innovation Scoreboard.

The report presents the most recent results for the IOI composite indicator and its underlying indicators for 46 countries, including EU Member States (MS) and selected EFTA, OECD and emerging economies. According to the IOI, Sweden, Ireland and Germany are the top-performing EU countries, followed by Denmark and Luxembourg. Sweden outperforms the other EU countries in patent applications, while Ireland and Luxemburg excel in employment within knowledge-intensive industries. Both Ireland and Germany achieve the highest performance in the domestic value-added content of their knowledge-intensive manufacturing exports, whereas Denmark records the highest domestic contribution to the value of its knowledge-intensive service industry exports. At the lower end of the ranking, Romania, Bulgaria, Slovakia, and Latvia exhibit the lowest performance among EU countries. Overall, most of the EU MSs (23 out of 27) improved their IOI performance between 2014 and 2024, with Lithuania and Estonia witnessing the most significant improvements during the period. The EU27 outperforms Canada, Australia, and China in the IOI, but it remains behind the United States, South Korea and Japan. Although the EU27's performance has improved between 2014 and 2024, this progress has occurred at a pace similar to or slower than that of some of its key partners. South Korea and Japan are either maintaining or widening their lead over the EU27, while China is narrowing the gap with the EU27. By contrast, the United States show signs of slower growth as the EU27 advances more rapidly.

The main IOI analysis is complemented by an exploration of innovation systems across MSs. This includes the collection and analysis of contextual indicators related to countries' economic structure and their performance in entrepreneurship and other dimensions of innovation. Additionally, a validation analysis of the IOI is conducted to assess its relationship with indicators reflecting inputs to the innovation process, framework conditions for robust innovation systems and its congruence with existing composite indicator frameworks measuring innovation at the country level. Overall, the analysis demonstrates a strong positive relationship between innovation performance, economic structure, and macroeconomic outcomes across EU countries. Top IOI performers such as Sweden, Germany, and Ireland report high GDP per capita growth, substantial R&D investment and a significant share of employment in innovation-active enterprises, while reflecting distinct economic structures. In contrast, low-performing countries like Bulgaria and Romania demonstrate improvements in both economic growth and IOI scores but continue to face limitations due to lower R&D intensity and a reliance on low-tech manufacturing.

The results of the IOI validation analysis demonstrate a strong positive correlation between the IOI and a range of indicators capturing innovation inputs and supportive framework conditions, as well as the rankings from other composite indicators monitoring innovation performance. While these overlaps confirm the IOI's alignment with established innovation metrics, the distinct differences in

rankings across frameworks underscore the added value of the IOI within the broader landscape of innovation monitoring tools.

Acknowledgements

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Authors

Michela Bello, European Commission, Joint Research Centre, Italy

Panagiotis Ravanos, European Commission, Joint Research Centre, Italy

Milene Tessarin, European Commission, Joint Research Centre, Italy

1. Introduction

The Innovation Output Indicator (IOI) is a composite indicator designed to support policy makers by providing an output-oriented metric of innovation performance at both national and European Union (EU) levels. The IOI was developed by the European Commission in response to the European Council's request to develop a "single integrated indicator to allow a better monitoring of progress in innovation" (European Council, 2011 and 2012). It captures a country's ability to translate innovation into economic benefits by examining how effectively innovative ideas reach the market, generate knowledge-intensive jobs, and enhance technological capabilities. By focusing exclusively on innovation outputs, the IOI monitors a streamlined set of dimensions of innovation, enabling policymakers to design or reinforce targeted actions to address innovation barriers and spur innovation performance.

The measurement framework of the IOI was introduced in the 2013 Communication and Staff Working Document (European Commission, 2013) and has been further refined in subsequent updates in 2014, 2016, 2017 and 2023. These refinements ensure the indicator remains aligned with the EU's evolving policy priorities.

The evidence provided by the IOI complements other benchmarking tools, such as the European Innovation Scoreboard (European Commission, 2025a), another tool produced by the European Commission, and it informs the biennial Science, Research and Innovation Performance of the EU (SRIP) report, which is the Commission's reference assessment of innovation performance of the European Union (European Commission, 2024b). In addition, the Commission adopted the IOI in the ex-post evaluation of Horizon 2020 to assess the progress of the program towards its objectives (European Commission, 2024a).

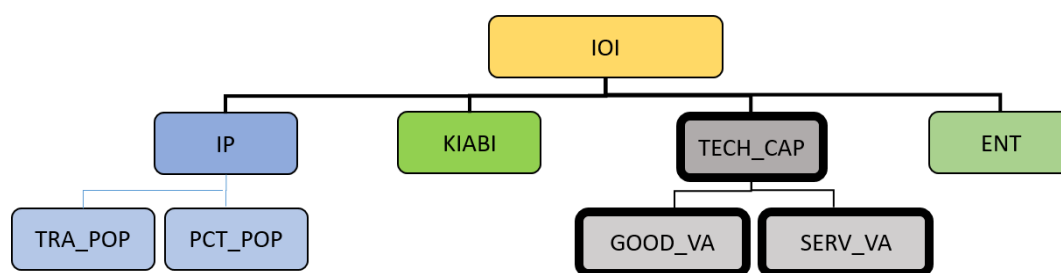
This report presents the latest results of the IOI covering the period 2014 and 2024. The current version retains the same analytical framework as the previous edition, comprising six indicators and four components. It incorporates updated data for most of the IOI's component indicators, covering 46 economies, including EU Member States (MS), the EU27 aggregate, and selected benchmark members of the EFTA, OECD and BRICS as well as additional emerging economies. The latest data used in this report refer to the year 2024; however, for certain indicators, the most recent available data refer to 2023 or earlier (see **Annex 5** for further details). The main IOI analysis is complemented by an exploration of innovation systems across MSs. This includes the collection and analysis of contextual indicators related to countries' economic structure and their performance in entrepreneurship and other dimensions of innovation. Additionally, a validation analysis of the IOI is conducted to assess its relationship with indicators reflecting inputs to the innovation process, framework conditions for robust innovation systems and its congruence with existing composite indicator frameworks measuring innovation at the country level.

The report is structured as follows. The next section outlines the conceptual framework of the IOI, detailing its structure and components. Section 3 analyses IOI performance across EU MSs and the EU27 aggregate, comparing them to benchmark countries, and examines performance variations across the IOI's four components. Section 4 provides a contextual analysis to deepen the understanding of IOI disparities among EU MSs, supported by validation tests assessing the robustness of the indicator. Section 5 concludes the report, summarizing key findings and policy implications. For completeness, **Annex 1** describes the dataset, methodological approaches, and procedures used to compute the IOI's aggregate measures. **Annex 2** outlines the robustness and sensitivity analyses conducted to evaluate the impact of methodological choices during the IOI's development.

2. Measuring country innovation performance: The IOI

The Innovation Output Indicator 2025 uses the revised IOI structure adopted in 2023 and used in the previous two editions of the index (Bello et al., 2023; 2024; 2025). (see **Annex 1**). The framework includes a total of six indicators aggregated into four components, including intellectual property (IP), knowledge-intensive activities in business industries (KIABI), domestic technology capacity (TECH_CAP), and innovative-active firms (ENT). These are depicted in Figure 1 and are described in more detail in **Annex 5**.

Figure 1. The Innovation Output Indicator Framework



Source: Authors' elaboration. Note: Indicator definitions are as follows, TRA_POP is trademark classes per million population, PCT_POP is patents filed under the patent cooperation treaty per million population, GOOD_VA is domestic value-added content of medium-high and high-tech manufacturing exports as a share of total manufacturing exports. SERV_VA is the domestic value-added content of knowledge-intensive service exports as a share of total service exports. ENT is the share of innovation-active firms on the total number of firms. KIABI measures the number of persons employed in knowledge-intensive business industries within total employment.

The IP component aims to capture a country's technological and non-technological innovation outputs. It consists of the arithmetic average of two indicators: the number of patent applications filed under the Patent Cooperation Treaty (PCT_POP) and the number of trademark classes in total resident trademark applications¹ (TRA_POP), each denominated by population (in millions).

The first indicator (PCT_POP) relies on patent data. It is well-known that patent data have a number of drawbacks and may not fully capture the whole set of innovation activities. Not all inventions, for instance, are patentable since they may not meet the patentability criteria (Dernis et al., 2001; Choi et al., 2007) or some organisations may prefer other mechanisms to protect their technologies (e.g., secrecy) (OECD, 2009). However, the use of patent information as a proxy of innovation output is well established in the innovation literature (Maurseth & Verspagen, 2002; Barzotto et al., 2019; Corradini et al., 2021) and they remain one of the most reliable indicators of innovating activity. The requirements of novelty, industrial applicability, and non-obviousness to deem an invention patentable have indeed been argued to be strongly related to innovative activity.

The limitations of patent data as a measure of innovation outputs are particularly evident in sectors with a low propensity to patent, such as services (Scherer, 1983; Arundel and Kabla, 1998; Brouwer

¹ A trademark application is filed by an applicant who can be resident or non-resident in the country of application. For example, a Belgian resident may want to protect their trademark in Germany (non-resident application). Each trademark application can cover multiple goods and services which are categorized into classes. Trademark classes are designed for international comparison and therefore suitable for the IOI.

and Kleinknecht, 1999). Following the 2023 framework revision, the IP component was introduced as a new addition to the IOI, comprising both patent data and a second indicator based on trademark data (TRA_POP). Information on trademarks offers another viewpoint on the multifaceted phenomenon of innovation, in particular one that is less focused on the technological side. Trademarks have been shown to convey information on non-technological innovations (Mendonça et al., 2004; Millot 2009), such as innovation in services or marketing innovation, while also being more closely linked to the market as they protect marketable products and support the commercialisation of innovation; moreover, they better capture innovative activity in small and medium-sized enterprises (SMEs) than traditional innovation indicators (Mendonça et al., 2004). The second component (KIABI) measures the number of persons employed in knowledge-intensive business industries as a share of total employment. It aims to capture the structural orientation of the business economy towards knowledge-intensive activities. This component has remained unchanged in the latest revision of the IOI, and it features in the index's previous editions (Vertesy and Damoli, 2020; Bello et al., 2024; Bello et al., 2025).

The third component, TECH_CAP, aims to capture the contribution of domestic technology capacity to the country's exports in knowledge-intensive manufacturing and service sectors. It is composed as the arithmetic average of two indicators: domestic value-added content of medium-high and high-tech manufacturing exports as a share of total manufacturing exports (GOOD_VA), and domestic value-added content of knowledge-intensive service exports as a share of total service exports (SERV_VA). The TECH_CAP indicator makes use of trade in value added data², which measures flows related to the value that is added by a country in the production of any good or service that is exported (OECD, 2013). Compared to similar metrics used in previous editions of the IOI, which make use of data on gross exports,³ the use of value-added data can differentiate between exports that reflect domestic technology capabilities and those that rely more on foreign inputs, such as those that are the result of mere imported components assembling. In addition, in a world of highly interconnected global value chains, traditional measures of trade based on gross exports face certain limitations as they record flows of good and services every time they cross borders and can lead to double counting and misguided policy decisions (OECD, 2013). The use of value-added data addresses this issue as a higher value in TECH_CAP indicates a lower foreign content of the country's exports in these industries.

The fourth and last component of the IOI is ENT, which is the share of innovation-active firms in the total number of firms. This indicator captures both product and business process innovations (i.e., process, marketing and organisation innovation), as well as other innovation activities by firms (e.g., R&D or training) which have not yet resulted in a product or a business process innovation. It also

² <https://www.oecd.org/sti/ind/measuring-trade-in-value-added.htm>.

³ This was the case for the precedent of the TECH_CAP component in previous editions of the IOI, namely competitiveness of knowledge-intensive goods and services (COMP), which was defined as the arithmetic average of two indicators, namely the share of medium and high-tech goods in total exports (GOOD) and the share of knowledge-intensive services in the total service exports (SERV). The COMP indicator was adopted in the IOI to measure the technological competitiveness of countries, i.e. their ability to commercialize the results of research and development (R&D) and innovation in international markets. It also reflects specialization in medium- and high-technology products, which have been found to be positively associated with economic growth, productivity and welfare, as well as with high value-added and well-paid employment (e.g. Hausmann et al 2007, Yoo 2008, Falk 2009).

reflects innovation activity across less knowledge-intensive economic sectors of a country, which are not captured by the KIABI or the TECH_CAP component.

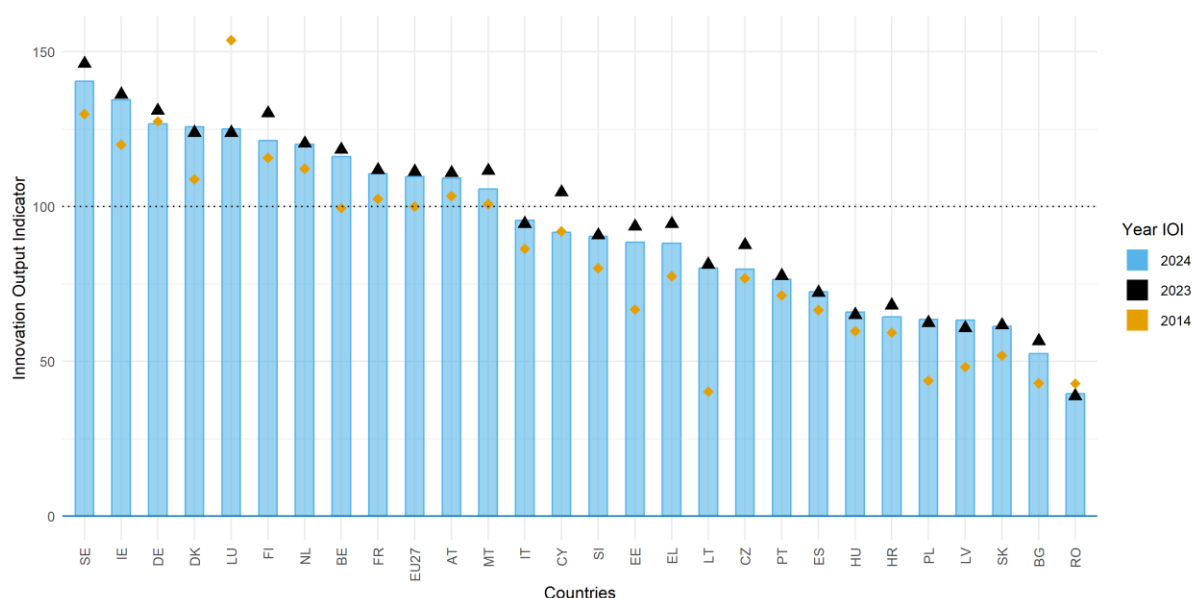
3. Results of the IOI 2025

3.1. Country performance in composite scores

This section presents the IOI 2025 scores obtained as the arithmetic average of the four components described in the previous section. The methodology for obtaining the IOI is the same as that of previous iterations and it is described in detail in **Annex 1**. **Figure 2** shows IOI scores in 2014, 2023 and 2024 for the 27 EU MS. The scores are normalised, with the EU27⁴ in 2014 set to 100, serving as the benchmark. Countries are ordered by means of their performance in 2024 (the blue bars).

Sweden is the country performing the best among the 27 EU MSs, followed by Ireland, Germany, Denmark and Luxembourg. On the opposite side of the ranking are Romania, Bulgaria, Slovakia, and Latvia. In total, nine of the 27 MSs Member States perform in 2024 better than the EU27. These include Western and Northern European countries (see the bars in **Figure 2**). On the other hand, Eastern European and Mediterranean countries are outperformed by the EU27.

Figure 2. IOI scores normalized to the EU27 score in 2014.



Source: Authors' calculation.

Between 2014 and 2024⁵, the IOI performance of 23 out of 27 EU Member States showed improvement, reflecting a positive trend in innovation output across the majority of the EU (see also **Table 10** in **Annex 4**). The largest improvements in the normalised IOI scores are observed in Lithuania (39.9 percentage points) and, though to a somewhat lesser extent, in Estonia, Poland,

⁴ In the text, we use the term "EU27" to refer to the European Union as a single entity (please see Box 1 for more details).

⁵ Given the number of methodological changes performed across the different editions of the IOI, the comparison of changes across years should be based on the latest version, which back casts IOI scores using the current methodology up to 2014, rather than using the previous IOI editions. See **Annex 1** for more details.

Denmark and Belgium (ranging between 21.7 to 16.6 percentage points). Between 2014 and 2024, IOI scores declined in four EU countries, with the most significant drop observed in Luxembourg, a moderate decline in Romania, and negligible changes in Germany and Cyprus.

The 2014–2024 trends suggest a convergence in innovation output performance between lower- and higher-performing EU countries, as low-performing economies advanced at a faster rate than their top-tier counterparts, narrowing the IOI gap over the decade. Particularly, the average change in the (normalised) IOI scores between 2014 and 2024 for MSs that were above the median in 2014 is 5.4% (the median is represented by Slovenia, for which the percentage increase in IOI is 12.8%). On the other hand, MSs below the median in 2014 exhibited on average an 21.9% change in their IOI with strong improvements in Lithuania, Estonia, and Poland.

Beyond the European MSs, Switzerland tops the IOI ranking for 2024, and as is illustrated in **Figure 3⁶**, had the same position in 2014 and 2023. It leads a cluster of countries with comparable IOI scores, namely Israel, the United States, South Korea, United Kingdom, and Japan. The EU27, follows leading a group of countries, which include Norway, Australia, Canada, China, and New Zealand, although differences in IOI scores are relatively small in the case of Canada and Australia. These are followed by a group of additional emerging economies with a markedly lower performance, namely Serbia, North Macedonia, Türkiye, Brazil, Mexico, and Chile. South Korea exhibited the largest absolute improvement in the IOI between 2014 and 2024 (from 110% of the EU27 score in 2014 to 134% in 2024), followed by China, North Macedonia, and United Kingdom.

Box 1. EU aggregate figures considered in the IOI report

All country and regional performance scores presented in this and the next section and in the report's Annexes benchmark country performance with respect to the performance of the EU27 as a single entity. For the IP component, and ENT and KIABI indicators, EU27 performance is constructed as the weighted average of the 27 country performances. This is equivalent to constructing, for example, the EU27 KIABI by adding the numerators and denominators of the 27 country KIABI ratios and then dividing them to obtain the EU27 aggregate score. The same is done for the TECH_VA indicators but with the difference that these exclude intra-EU trade to consider the EU27 as a single entity trading with the rest of the world. This measure (denoted as EU27 and used as such throughout the report) is used to compare the IOI scores of the EU as a single entity with non-EU economies. Its use is in line with the benchmarking done in other monitoring frameworks such as the European and the Regional Innovation Scoreboard.

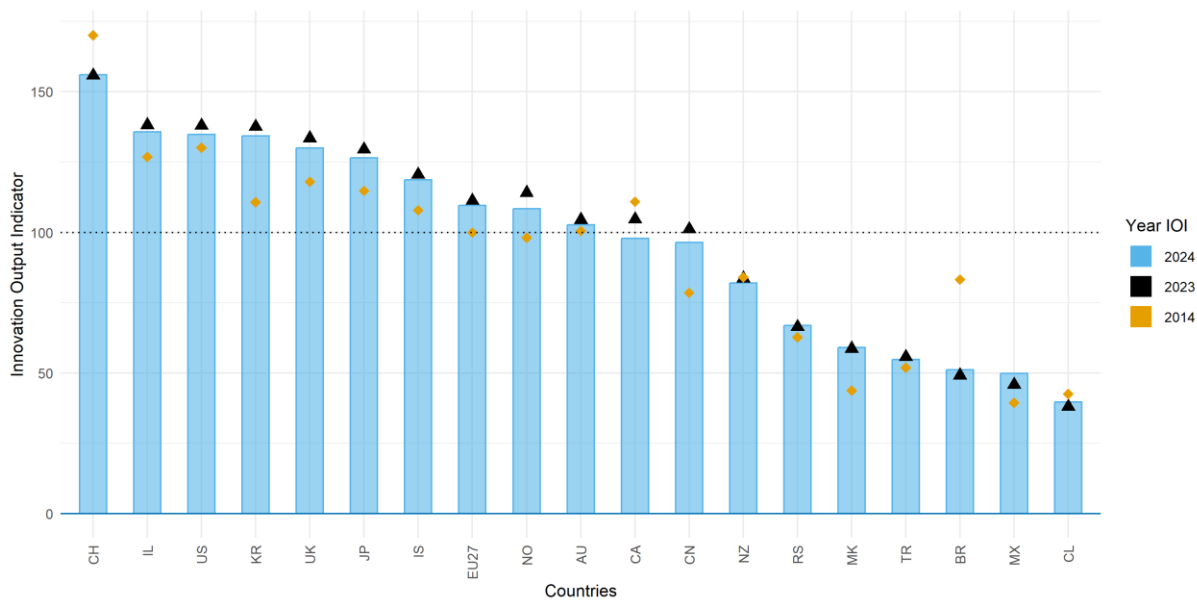
In the figures depicting individual indicators shown in Section 3.2, the simple, unweighted, average of the 27 EU countries (denoted as "EU Average") is also presented, to facilitate a comparison of the performance of individual EU MSs with that of the "average EU country". In this context, the EU average of TECH_VA estimates include both extra and intra-EU trade as it is constructed as a simple, unweighted average of country-level TECH_VA values.

The EU27 exhibited an improvement in innovation output performance as measured by the IOI of about 9.6% (its IOI is 109.6% of its 2014 value). This is similar to that of Japan and United Kingdom (10.1% and 10.2%, respectively, relative to the EU27 IOI in 2014), but it falls short of the improvements in South Korea (21.3%, from 110.8 to 134.3 in 2024) and China (22.8%, from 78.5 in

⁶ The EU27 benchmark excludes intra-EU trade, making it suitable to compare scores of the EU as a single entity with non-EU countries (see Box 1).

2014 to 96.4 in 2024). It is however larger than the improvement of the US (3.6%, from 130.1 in 2014 to 134.8 in 2024), suggesting that the growth of EU27 between 2014 and 2024 has slightly surpassed that of the US. On the other hand, Brazil exhibited the most marked performance decline (-38.5%), which arises from a decline in its performance across various IOI indicators especially in the context of innovation-active enterprises, and, to a lesser extent, patent applications, domestic value added in medium-high and high-tech manufacturing and service exports. Switzerland exhibited a sizable performance decline of 8.3%, owed mostly to a 28% decline in the share of innovation-active enterprises, and Canada also presented a 11.8% performance decline, mostly due to a decline in the trademark applications. The remaining countries exhibited smaller variations in their performance in the 2014-2024 period.

Figure 3. Comparison with world / benchmark countries: IOI score normalized to the EU27 score in 2014.



Source: Authors' calculation. Note: Horizontal line shows EU performance in 2014 (normalised to 100).

The performance scores for non-EU countries should be read with some caution. Despite the use of matching guidelines provided by, for example, Eurostat for the US and Japan⁷, differences in industrial classification and coverage may imply that KIABI scores are not fully comparable between non-EU countries and the EU27. Moreover, the presence of missing indicators is somewhat more pronounced for non-European countries, in particular for North Macedonia and Serbia (GOOD_VA and SERV_VA), China (KIABI), Canada, Israel and Mexico (ENT).⁸ For comparisons across time, differences in data source (OECD, Eurostat and National Offices) within the same country are present in a few countries for ENT (Switzerland, United Kingdom) and KIABI (Japan, North Macedonia, South Korea, Türkiye, United Kingdom).

⁷ See https://ec.europa.eu/eurostat/cache/metadata/Annexes/htec_esms_an_8.pdf

⁸ See **Annex 1** for details on how data are imputed to build the final IOI in these cases.

3.2. Analysis by indicator

The IOI is a composite indicator that captures the general trend of a country's performance by aggregating the six underlying indicators that measure different aspects of innovation. As such, it provides a concise *summary* of multidimensional performance in terms of innovation output. However, to have a more nuanced understanding of the factors driving strong or weak country performance in the IOI, it is necessary to examine trends and performance at the indicator level. This will identify country strengths and bottlenecks in the individual indicators comprising the IOI.

This subsection discusses country performance for each of the six indicators that comprise the IOI. The six figures below depict performance trends for the years 2014, 2023, and 2024 for the 46 economies included in the IOI.⁹ The figures also enable comparison of the 27 EU countries (light blue bars) with the “average EU country” (dark blue bar) (See **Box 1**). Alternatively, the performance of the EU27 as a single entity (gold bar) can be compared to that of non-EU countries (coral-coloured bars).

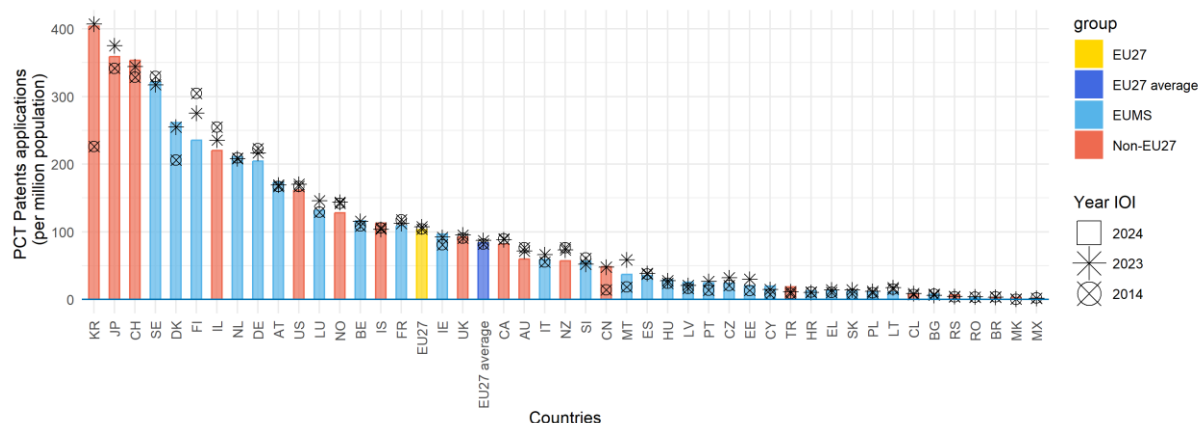
South Korea leads the 2024 ranking in terms of per-capita PCT patent applications (PCT_POP), followed by Japan, Switzerland, and Sweden (**Figure 4**).¹⁰ These four countries have registered more than 320 patent applications per million citizens in 2024. Mexico, North Macedonia, and Brazil lag at the tail of the distribution having registered less than 3 PCT patent applications (per million population). With regards to changes between 2014 and 2024, the majority of the countries included in the IOI (29 out of 45) exhibited an increase in patent activity per million population, although the EU average has remained stable over this period. China and Türkiye more than doubled their per capita patents between 2014 and 2024.¹¹ A group of EU countries (Portugal, Latvia, Estonia, and Slovakia) exhibited marked increases ranging from 40% to 80%, although their performance in 2024 is still below the average EU country. The patenting activity of the EU27 during this decade has increased slightly to about 109 patents per million population in 2020 before returning to 103 patents per million population in 2024 (which is equal to the 2014 level). This decline may reflect pandemic-related effects, including, for instance, economic uncertainty and weaker or disrupted R&D activity during the COVID-19, although other structural or cyclical factors may also have contributed to this pattern.). These figures lag behind those of EU's main competitions, such as South Korea, the US and Japan.

⁹ JRC collects data for additional European, non-EU countries, which currently do not meet the inclusion criteria for the IOI (see **Annex 1**) but are considered potential candidates for inclusion in future editions of the IOI. Currently, six additional countries (Albania-AL, Armenia-AM, Bosnia and Herzegovina-BA, Georgia-GE, Moldova-MD, and Montenegro-ME) are being monitored.

¹⁰ The years in the Figures refer to the IOI reference period (Year IOI) and correspond to a zero, one, or two-year shift relative to the years of the observed data due to carrying forward of missing observations for the most recent years (see **Annex 5**). Particularly for patent applications, the final data point in our data series is 2022, accounting for typical delays in patent data collection and updates

¹¹ The same is true for a group of small countries (North Macedonia, Cyprus, Malta). These large increases should however be viewed with some caution as the patent figures for these countries were very low in the beginning of the period covered by the study (less than 10 patents), while their trend is also volatile over time (for Malta and North Macedonia).

Figure 4. Number of patent applications filed under the Patent Cooperation Treaty per million in population (PCT_POP)



Source: See **Annex 5**.

Figure 5 depicts the number of trademark classes¹² of resident applications per capita (TRA_POP) across countries, with the exception of Greece and Ireland for which this indicator is not available¹³. In 2024, Malta leads the ranking, followed by Luxembourg which ranks second in both 2024 and 2023 but topped the ranking in 2014.¹⁴ These are followed by a group of non-EU countries (South Korea, China, Switzerland, Türkiye, and Iceland), among which China exhibited the largest improvement between 2014 and 2024, more than tripling its trademark class registrations per million population. South Korea and Türkiye almost doubled their per capita trademark class registrations. . Good performance in this indicator could also be linked to the presence of more favourable IPR-related regulations and procedures, firms' strategies, and the relative importance of certain economic sectors. For example, high propensity sectors for trademarks like business services and advertisement have a substantial share in Luxembourg, while the gaming and software sectors are relevant in Malta (European Commission, 2022).¹⁵

At the opposite side of the 2024 ranking Israel, Serbia, and North Macedonia show the lowest performance, with less than 1000 trademark classes per capita. North Macedonia reduced its trademark registration per capita by 26% compared to 2014, while Israel remained roughly unchanged, and Serbia improved in relation to its 2014 performance (13%). Trademark registrations for the EU27 as a whole increased by 4.3%, reaching about 2590 trademarks per capita in 2024, well above some of its competitions, such as the US and Japan.

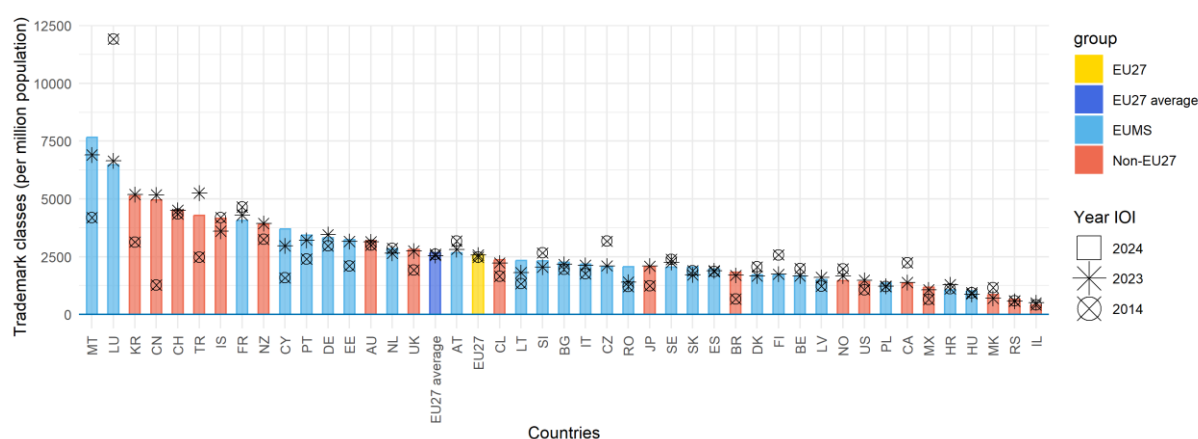
¹² Trademarks are registered with specific class codes – one for each type of product or service the trademark applies to. A single trademark application can cover many trademark classes.

¹³ **Annex 1** describes how data are imputed to build the final IOI for these countries.

¹⁴ Luxembourg reached their highest point in 2016 with about 13 thousand trademarks per million population. Ever since there is a continuous decrease in this indicator until 2023 (the latest available year), which is driven primarily from the decline in registered trademarks (-43% from 2016) rather than the population increase in the country (14.5% from 2016).

¹⁵ Country size is also a partial contributor to the relatively better ranking of countries like Luxembourg and Malta. The indicator is expressed per capita and thus larger countries tend to be relatively disadvantaged in the ranking, since higher population levels dilute otherwise high absolute volumes of trademark activity.

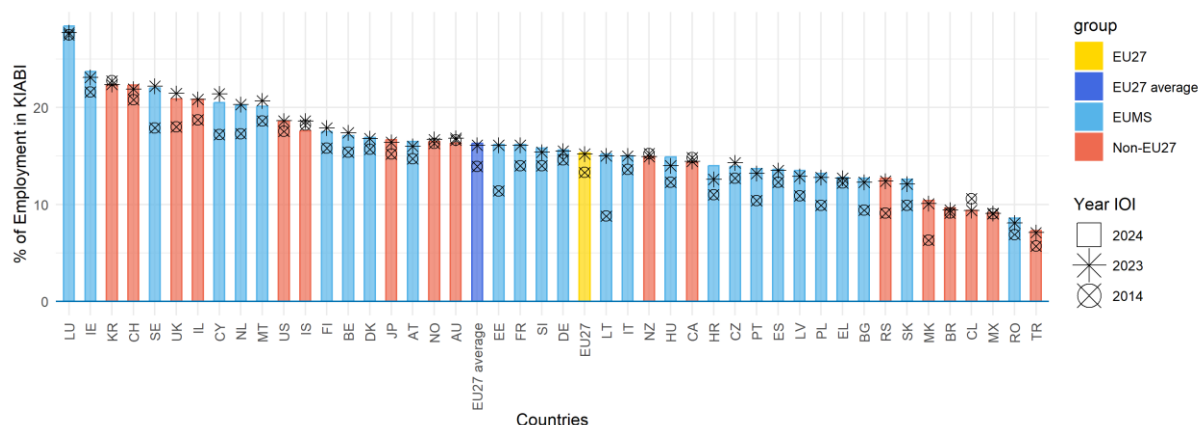
Figure 5. Number of trademark classes of resident applications per million in population (TRA_POP)



Source: See **Annex 5**.

Figure 6 shows the share of employment in knowledge-intensive activities in business in total employment industries (KIABI). For China this indicator is not available. Performance on this indicator relates to the structure of a country's economy. Therefore, countries with strong financial and/or ICT service sectors tend to perform better than the rest. Luxembourg tops the ranking across the whole decade from 2014 to 2024, with a lead (28.4% in 2024) over Ireland, South Korea, Switzerland, and Sweden, which rank in the 2nd to 5th positions (ranging from 22% to 23.7%). Türkiye, Romania, and Mexico, are at the bottom of the ranking for 2024, having less than 10% of their economy employment occupied in jobs in knowledge-intensive sectors.

Figure 6. Employment in knowledge intensive activities in business industries (KIABI) as percentage of total employment



Source: See **Annex 5**.

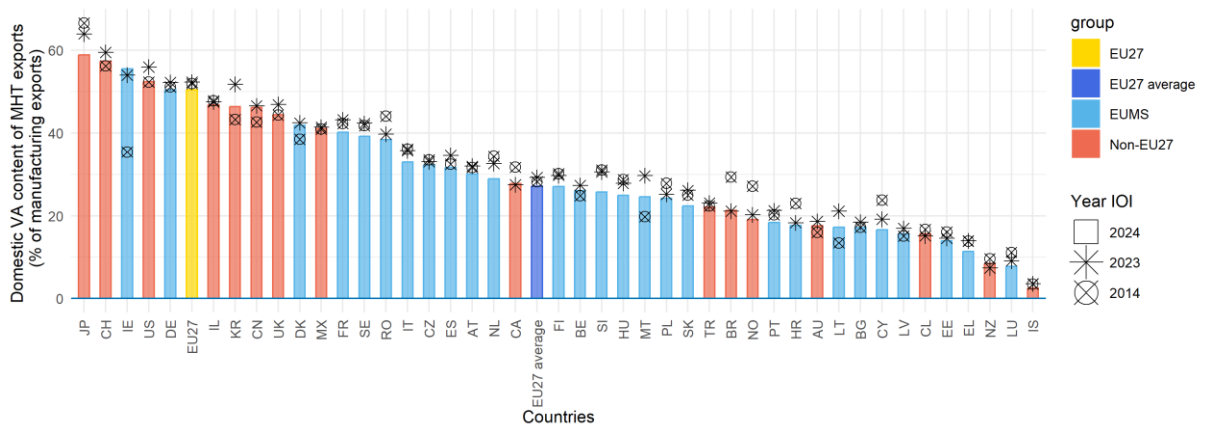
KIABI has been growing in the vast majority of the considered countries (38 out of 45) between 2014 and 2024. Lithuania, Estonia, North Macedonia and Sweden exhibited the largest improvements, increasing their share of employees in knowledge-intensive activities in business industries by more than 4 percentage points in the last decade. Fourteen other countries recorded a growth of 2 to 4 percentage points during this decade, while another 20 countries exhibited marginal improvements of less than 2 percentage points. In contrast, KIABI declined marginally (1 percentage point or less) in six countries, all of which are non-EU members (Iceland, Canada, South Korea, New Zealand, Australia,

and Chile) For the EU27, the gradual shift of employment towards knowledge-intensive activities is a recognised megatrend, particularly for services (see, e.g., Cedefop, 2021). Between 2014 and 2024, KIABI increased for all EU MSs (ranging from an increase of 6.4 percentage points in Lithuania to 0.6 in Greece), while the EU27 exhibited an increase of 2.4 percentage points, achieving 15.3% in 2024.

Figure 7 and **Figure 8** depict the domestic value-added content of exports, respectively in high and medium-high technology manufacturing (as a percentage of total manufacturing exports) and knowledge-intensive service industries (as a percentage of total service exports). Data on these indicators are not available for North Macedonia and Serbia (see **Annex 3** for more details).

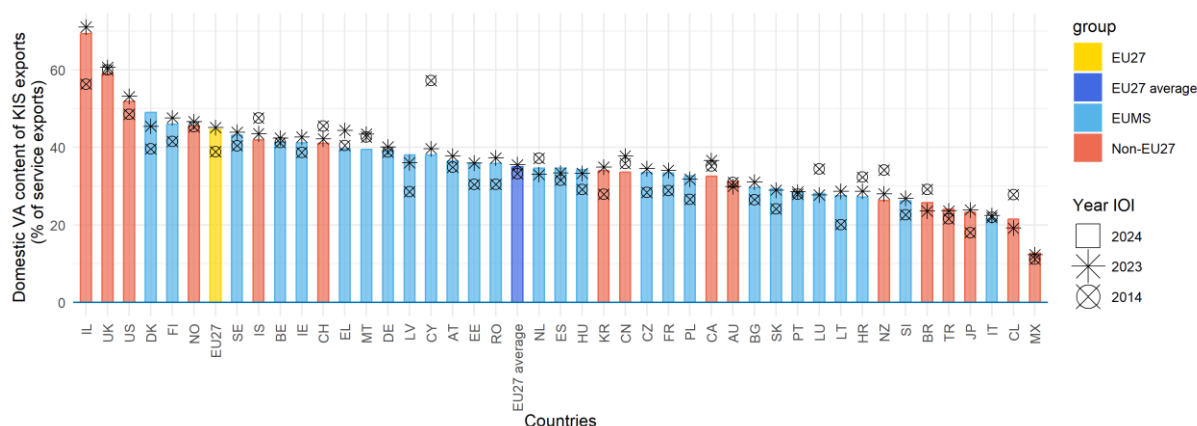
Japan is the country with the highest domestic contribution to the value of its high and medium-high technology manufacturing exports (58.9% in 2024) for all years between 2014 and 2024. It is followed by Switzerland, Ireland, the United States, and Germany, with a share above 50% in the manufacturing sector (GOOD_VA). This reveals the lower dependency on foreign inputs in knowledge-intensive exports for these countries, for all of which more than 50% of the value added in knowledge-intensive manufacturing exports is created by the domestic industry. In terms of services, Israel is the top performer in 2024 with 69.3% of the value of their knowledge-intensive service exports added from domestic industries, followed by the United Kingdom and the United States with more than 50%. These top performers are followed by eight other countries (five of which are EU members) for which SERV_VA is between 40% and 50%: Denmark, Finland, Norway, Sweden, Iceland, Belgium, Ireland, and Switzerland. Notably, Japan ranks among the countries with the lowest performance in terms of domestic value added to service exports. On the opposite side of the 2024 rankings, Iceland, Luxembourg, and New Zealand show the lowest domestic contribution to exports in high and medium-high tech manufacturing industries (less than 9%), and Italy, Chile, and Mexico do so in knowledge-intensive service exports (with less than 21%).

Figure 7. Domestic value-added content of exports in high and medium-high technology manufacturing (MHT) industries as percentage of total manufacturing exports



Source: See **Annex 5**.

Figure 8. Domestic value-added content of exports in knowledge-intensive service industries as percentage of total service exports



Source: See **Annex 5**.

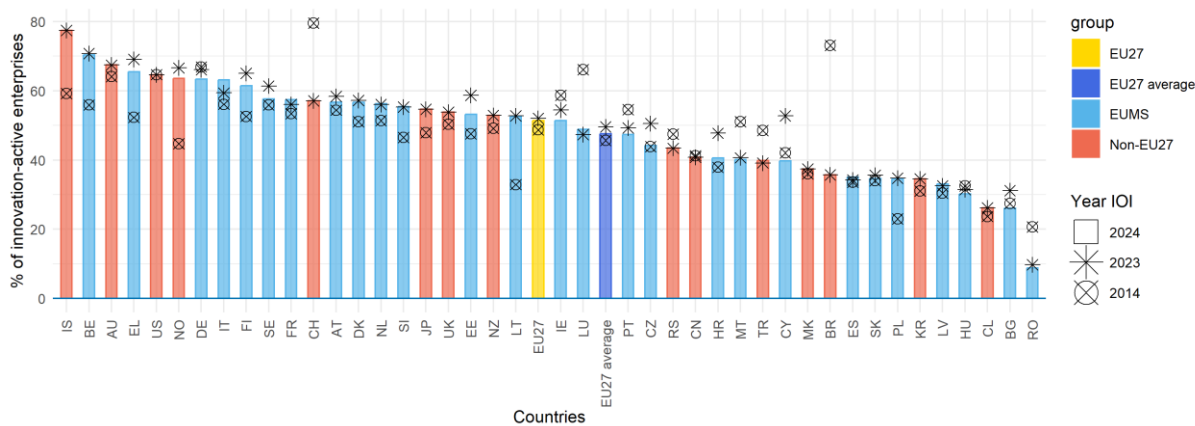
Between 2014 and 2024 most of the considered countries exhibited a drop in the domestic contribution to the value added of their knowledge-intensive manufacturing industries (30 out of 43 countries). The share of domestically added value declined by 8 percentage points in Brazil and Norway, while Slovenia, Hungary, Netherlands, Romania, Cyprus, and Japan experienced a drop of at least 5 percentage points in the decade from 2014 to 2024. In contrast, the domestic contribution to knowledge-intensive exports in services industries has risen in roughly two-thirds of the countries analysed (28 out of 43). Overall, the average increase was more modest in terms of the domestic value added in manufacturing exports than in service exports. Ireland experienced the largest improvement in manufacturing industries, while Israel showed the most significant improvement in services, with an increase in the domestic value added of its exports of more than 10 percentage points. In turn, Cyprus exhibited a marked decline in the share of domestic value added in knowledge-intensive service exports of 19 percentage points, followed by Chile, Luxembourg and New Zealand, with declines of 6 to 8 percentage points, and 11 other countries with less significant declines.

Overall, the EU27 relies less on foreign inputs in its knowledge-intensive manufacturing exports (50.4% domestic contribution in 2024) compared to its knowledge-intensive service exports (44.5% domestic contribution in 2024). The EU27 has increased its domestic contribution in the latter over the last decade by 2.6 percentage point, but it recorded a marginal decrease of 1.5 percentage points in the former between 2014 and 2024. The EU27's domestic contribution to knowledge-intensive manufacturing exports in 2024 exceeds that of South Korea, China, and the United Kingdom, and it is closely following the United States. In the service sector, EU27 lags behind Israel, the UK, and the USA, but performs better than China and South Korea. It is also noteworthy that the scores for the EU27 as a single entity are significantly higher than those for the EU average, highlighting a substantial integration of EU Member States on intra-EU trade.

Figure 9 presents the percentage of innovation-active enterprises, with the exception of Canada, Israel, and Mexico, for which this data is not available. In 2024, Iceland recorded the highest proportion of enterprises engaged in innovation activities (77.4%), followed by Belgium, Australia, Greece, and the United States, all exceeding 63%. In contrast, Romania reported the lowest share (8.8%), with Bulgaria, Chile, and Hungary closely following at 26–30%. As shown in the figure, this indicator exhibits a high volatility over the 2014-2024 period. While five countries experienced a significant decline (more than 10 percentage points) in the share of innovation-active enterprises during this decade, another six countries saw an increase of more than 10 percentage points. This

volatility reflects the dynamic nature of business activity, as well as potential methodological changes in national surveys. Substantial improvements of over 15 percentage points were observed in Lithuania, Norway and Iceland, whereas a decline of more than 37 percentage points occurred in Brazil, and 17 to 22 percentage points in Switzerland and Luxembourg. For the EU27 group, the share of enterprises engaging in innovation activities has remained relatively stable over time, with a slight increase of 2.5 percentage points (to 51.4%) between 2014 and 2024.

Figure 9. Percentage of innovation-active enterprises as percentage of all enterprises



Source: See **Annex 5**.

The analysis above highlights a relatively high heterogeneity in country performance across the various dimensions of innovation included in the IOI. This variability within the components of a composite index often serves as evidence of the added value of integrating these dimensions into a single measure of the socio-economic phenomenon under study (in this case, innovation output activity). By combining multiple indicators, the composite framework not only summarises overall trends but also reveals aspects of the phenomenon that may not be immediately apparent when analysing individual indicators in isolation (Saisana *et. al.*, 2022). This also holds for identifying the sources of differences in the IOI across countries and between EU27 and its main economic partners (see Box 2).

For example, within the EU, Sweden is the top performer in PCT patent applications per million in population, and it is third in terms of employment in knowledge intensive activities in business industries, while it performs relatively worse in terms of trademark applications. Belgium performs relatively better in terms of share of innovation-active enterprises than in terms of trademark and patent applications. Ireland ranks 3rd in terms of domestic value added to its high and medium-high tech manufacturing exports and 2nd in KIABI, but only 18th in terms of patent applications.

A comparable degree of heterogeneity is also observed among the EU’s international partners. Switzerland ranks among the top five countries in all IOI indicators with the exception of domestic value-added content of knowledge-intensive service exports. South Korea demonstrates strong performance in patent and trademark applications, as well as in the share of employment in knowledge-intensive activities within business industries but does not exhibit a similarly strong position in ENT. Japan holds a leading position in patent applications – though not in trademark applications – and in domestic value added of high and medium-high-technology manufacturing exports, but not in service exports.

Box 2. Analysing the performance gap between EU and main economic partners

The analysis of individual indicators also provides further insight into the sources of performance gap between the EU27 and its main international economic partners. Comparing EU27 performance (gold bar) with that of non-EU countries (coral-coloured bars) suggests that the EU27's underperformance is most pronounced in the following indicators:

- The EU27 share of employment in knowledge-intensive business industries is outranked by the US, Japan, and the UK and lags almost seven percentage points behind South Korea and Switzerland (roughly 15% versus 22%). However, as previously noted, the KIABI has increased in the EU27, approaching that of Japan and of the US.
- Patents per capita in the EU27 lag considerably behind those of Korea, Japan, and the United States, with differences ranging from 60 to 300 patents per capita. In contrast, EU27 outperforms US and Japan and is roughly at par with the UK in terms of trademarks per capita but still trails considerably behind South Korea.
- Despite EU27 outperforming South Korea and being comparable with the UK and Japan in the share of innovation-active enterprise, it still lags significantly behind the US, where 64.7% of enterprises are innovation-active compared to 51.4% in the EU27.

Over the last decade, China has made significantly faster progress than the EU in most IOI indicators, underscoring its rapid emergence as a major player in the global innovation landscape. For instance, China improved in most of the IOI indicators between 2014 and 2024, except in the case of SERV_VA, where it dropped nine positions, while, on the other hand, it exhibited a marked improvement in trademark applications (from 35th to 4th in the ranking). China's dependence on foreign inputs in knowledge-intensive exports has increased, as reflected by a 2.2 percentage point decline in its SERV_VA share. In contrast, the EU27 has reduced its reliance on foreign inputs in the production of knowledge-intensive services over the past decade, as evidenced by a 5.7 percentage point increase in its SERV_VA share compared to 2014.

On the other hand, progress in the US over the last decade (2014-2024) has been more modest – both in absolute and relative terms – compared to the EU for most of the IOI indicators, giving rise to the more nuanced growth of the US with respect to the EU27 in terms of the IOI documented above. For example, KIABI in the US increased by 1 percentage point, whereas it rose by 2 percentage points in the EU27. Similarly, EU27 increased the share of value added to its knowledge intensive service exports by about 5.7 percentage points, roughly two times as much as the corresponding increase in the US (3.3). Furthermore, the EU managed to maintain a stable production of patents per capita between 2014 and 2024, while the US exhibited a decline in their production by 3.2% (or roughly 5.3 patents per capita).

4. Additional analysis

4.1. Contextual indicators

To enhance our understanding of the differences across EU countries regarding the IOI, as well as its individual components, a set of contextual indicators is provided in **Annex 6**. These selected indicators include macroeconomic dimensions, such as economic performance, structure of the economy, entrepreneurship, and innovation.

The top-performing countries in the IOI, notably Sweden, Ireland, and Germany, also ranked among the top 10 in GDP per capita (see **Table 11** in **Annex 6**) in 2024. They exhibited a consistent GDP growth rate exceeding 30% over the period from 2014 to 2024, thereby confirming the positive association between innovation and economic performance. However, these countries differ in terms of their economic structure. Germany boasts one of the highest shares of employment in the manufacturing sector among EU countries. Despite the natural decline in the manufacturing share within the total economy that high-income countries tend to experience (Tregenna, 2016), Germany has managed to keep the decline relatively low over the last decade, with a 7.1% reduction between 2014 and 2024. Conversely, Sweden and Ireland are more service-oriented economies, with 81.0% and 77.6% of total employment in service activities, respectively (ranking among the top ten EU countries). They fall below the EU average of 15.3% for manufacturing employment, with Sweden at 9.5% and Ireland at 10.6% (see **Table 11**). Interestingly, Sweden and Ireland also hold more than half of their service employment to knowledge-intensive sectors (**Table 12**), which exhibit a higher propensity for innovation.

The productive capacities of these countries also reflect well in terms of innovation output. Germany holds the 2nd position in the IOI component that represents the contribution of domestic technological capacity to the country's exports in high and medium-high manufacturing (GOOD_VA). Meanwhile, Sweden and Ireland are among the leading performers in terms of domestic value-added content in knowledge-intensive service exports (SERV_VA) within the IOI. In contrast, Bulgaria and Romania lie at the opposite end of the IOI ranking. Both countries rank among the lowest in GDP per capita within the EU (**Table 11**). Nonetheless, Bulgaria experienced substantial GDP per capita growth over the period, with a 105.5% increase from 2014 to 2024. Concurrently, Bulgaria improved its IOI index compared to 2014. Similarly, other countries, including Latvia (70.7%) and Slovakia (43.5%), demonstrated significant GDP per capita growth rates between 2014 and 2024 and an improvement in the IOI index between the same period. Bulgaria and Romania share another common characteristic: they record the lowest share of employment in the service sector among EU countries (**Table 12**). Additionally, although they exceed the EU share in terms of manufacturing employment, most of these jobs are concentrated in low and medium-low tech sectors, which tend to yield fewer innovative outcomes – as shown in **Table 12**. In 2024, Bulgaria had 4% employment in high and medium-high tech manufacturing sectors in 2024, compared to 6.9% in Romania.

Enterprise birth rates exhibit considerable variation across EU countries (see **Table 13**). In 2023, these rates ranged from 6.2% in Austria, 7.3% in Denmark, 7.8% in Italy, and 8.2% in Sweden to almost 20% in Lithuania, 17.1% in Malta, and 16.8% in Portugal. Moreover, Lithuania, Malta, and

Portugal display relatively high shares of employment in fast-growing enterprises¹⁶, ranging from 16% to 25% of total employment, surpassing the EU average (12.9%) in 2023. As observed in the IOI components, Malta and Portugal rank among the top five EU countries in trademark applications per million population, an indicator more closely related to the creation of new enterprises and non-technological innovation. It is noteworthy that none of the top innovators measured by the IOI possess high enterprise birth rates. Overall, 16 EU countries witnessed a positive growth rate in enterprise birth from 2021 to 2023, but at the EU level it remains slightly negative at -3.1% during this period (**Table 13**). Enterprise birth rates may vary significantly across sectors and in response to support packages aimed at encouraging entrepreneurship post-COVID pandemic or demographic changes. Over the years, there has been a steady decline in business dynamics within the EU, raising concerns about the implications for innovation, as enterprise births are believed to enhance competitiveness, stimulate innovation, and facilitate the adoption of new technologies (European Commission, 2022).

Analysis of innovation inputs (**Table 14**) reveals a clear correlation between high investment and strong performance, with Sweden, Germany, and Denmark serving as notable examples of EU MSs that achieve the highest results through substantial innovation efforts. Leading the IOI ranking are countries that allocate substantial resources to R&D investments, both in terms of volume and percentage of GDP. Additionally, these countries boast the highest shares of employment in innovation-active firms. In 2024, Sweden dedicated 3.4% of its GDP to R&D activities, while Denmark invested 3.0%. Germany accounted for nearly one-third of the entire EU's R&D investment in 2024 (in purchasing power parity, at 2005 constant prices). Sweden, Ireland, and Germany, the top three countries in the IOI, had over 70% of their total employment within innovation-active firms, as a proportion of the countries' total employment. Furthermore, Ireland and Sweden excelled in the IOI component pertaining to employment in knowledge-intensive activities within business industries (KIABI).

Conversely, Romania, Bulgaria, and Slovakia, which recorded the lowest performance in the IOI, had relatively low R&D investment rates in 2024, ranging between 0.5% and 1% of GDP, among the lowest in the EU (**Table 14**). Romania has only 19.3% employment in innovation-active firms, while Bulgaria and Slovakia perform better, with employment shares in innovation-active firms surpassing 50% in relation to total employment (51.8% and 57.1%, respectively).

4.2. Validation analysis

As a composite indicator, the IOI summarizes a latent concept that cannot be directly measurable by a single metric. The validity of the IOI as a measure of innovation can be assessed by examining its concordance with alternative measures of national innovation performance or comparable constructs for the same countries, as well as other unidimensional indicators that resemble or are closely related to innovation output, such as inputs to innovation and framework conditions known to facilitate the process of innovation. This subsection summarises the results of such a validation analysis (see Annex 2 for a more detailed discussion).

The IOI is positively associated with of two other summary measures of innovation at the national level, namely the 2025 Summary Index of the EIS (European Commission, 2025a) and the WIPO Global Innovation Index (WIPO, 2025) and their sub-components (see **Table 7** and **Figure 15, Figure**

¹⁶ Fast-growth is defined by an annual average employment growth greater than 10% over a three-year period.

16, and **Figure 17** in Annex 2). The fact that leader and laggards in the IOI are similarly performing in the EIS and GII supports the consistency of the IOI, indicating that it indeed measures a similar phenomenon to these metrics. However, notable differences are observed in the relative performance of countries across the three measures (see **Table 7** in Annex 2), suggesting that their close association does not indicate information redundancy. On the contrary, they suggest that the IOI provides a distinct perspective on country innovation performance relative to the EIS and GII.

On the other hand, the IOI and its four components are adequately correlated with a series of variables representing inputs to innovation, such as R&D expenditure and GDP per capita, and the quality of innovation-enabling framework conditions within countries, such as the share of higher education graduates, ICT specialists, and cloud computing adoption within the country (see **Table 8** in Annex 2). This provides additional evidence on the validity of the IOI, in that countries investing more in innovation and facilitating a more innovation-friendly environment tend to also perform better in terms of the innovation outputs captured by the IOI.

5. Conclusions

This report presents the 2025 edition of the Innovation Output Indicator (IOI), an output-oriented measure of innovation performance at the country and EU levels. The IOI assesses countries' capacity to derive economic benefits from innovation by tracking the extent to which innovative ideas reach the market, create knowledge-intensive jobs, and increase country technological capability.

According to the IOI¹⁷, Sweden, Ireland, and Germany are the best-performing EU MSs in 2024 and are followed by Denmark and Luxembourg. Sweden outperforms the other EU countries in terms of patent applications; Ireland does so in terms of employment in knowledge intensive industries and also, together with Germany, outperform the other EU MSs in terms of their domestic value-added content of their knowledge-intensive manufacturing exports. Luxembourg tops the ranking of trademark applications and shares of employment in knowledge-intensive activities, while Denmark excels in the value of knowledge-intensive service exports added from domestic industries and patent applications.

The EU27 leads international partners such as Canada, Australia, and China, but it continues to lag behind the United States, South Korea and Japan. These results are mainly due to the EU27's relatively weaker performance in terms of the share of knowledge-intensive employment in business industries, patent and trademark applications, and the share of innovation-active enterprises. South Korea and Japan are either maintaining or widening their lead over the EU27, while China is narrowing the gap with the EU27. By contrast, the United States show signs of slower growth as the EU27 advances more rapidly.

Section 4 compares the IOI with other innovation-related indicators and frameworks at the country level to validate its relationship with innovation inputs, framework conditions, and broader contextual measures. Higher innovation output in the EU is systematically linked to higher income levels, sustained economic growth, and greater R&D investment. Conversely, countries with lower IOI performance are typically characterised by lower R&D intensity and a reliance on low-tech manufacturing.

The IOI reveals a robust positive association between national performance in the IOI and indicators of innovation inputs and enabling conditions. Although the IOI demonstrates meaningful correlation with the GII, the European Innovation Scoreboard (SII), and their components, the substantial rank discrepancies between the IOI and these indices underscore its capacity to capture dimensions of country innovation output that other frameworks do not address.

¹⁷ As noted on p. 9, the 2025 IOI edition presents data from the previous year (2024) or the most recent year available.

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List of country abbreviations and definitions

Code	Country	Code	Country
AT	Austria	TR	Türkiye
AU	Australia	UK	United Kingdom
BE	Belgium	US	United States
BG	Bulgaria		
BR	Brazil		
CA	Canada		
CH	Switzerland		
CN	China		
CL	Chile		
CY	Cyprus		
CZ	Czech Republic		
DE	Germany		
DK	Denmark		
EE	Estonia		
EL	Greece		
ES	Spain		
EU27	European Union		
FI	Finland		
FR	France		
HR	Croatia		
HU	Hungary		
IE	Ireland		
IL	Israel		
IS	Iceland		
IT	Italy		
JP	Japan		
KR	South Korea		
LT	Lithuania		
LU	Luxembourg		
LV	Latvia		
MK	North Macedonia		
MT	Malta		
MX	Mexico		
NL	Netherlands		
NO	Norway		
NZ	New Zealand		
PL	Poland		
PT	Portugal		
RO	Romania		
RS	Serbia		
SE	Sweden		
SI	Slovenia		
SK	Slovakia		

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Annexes

Annex 1. The IOI methodology

The IOI 2025 dataset and index development¹⁸

The IOI 2025 dataset includes 506¹⁹ country-year observations covering the 27 European Union MSs, EFTA countries (Switzerland, Iceland, Norway), EU candidates (North Macedonia, Serbia, Türkiye), selected OECD countries (Australia, Canada, Chile, Israel, Japan, South Korea, Mexico, New Zealand, United Kingdom, United States of America) and two BRICS members (Brazil and China) over the time period 2014 until 2024. It also includes an observation for the 27 EU MS as a single economic zone (EU27). The dataset includes six indicators: PCT_POP, TRA_POP, KIABI, GOOD_VA, SERV_VA and ENT. JRC conducted data collection for additional countries. The threshold for inclusion of a country in the IOI has been set to 65% of the indicators having at least one non-missing data point, as per the JRC guidelines (JRC/OECD, 2008). That is, countries are included in the IOI calculations if they have at least one non-missing data point for four out of the six indicators. Countries that did not meet this criterion for which data were collected (Albania, Armenia, Bosnia and Herzegovina, Georgia, Moldova, Montenegro) are included in the indicator Figures in Section 3 whenever data for those countries are available.

Comparison with earlier IOI reports

It must be stressed that comparisons with results from previous IOI reports are not possible. Although the conceptual framework and the methodology in the current report is the same as in the previous three IOI reports, results for the same year, e.g. 2021 in the IOI 2024 report (Bello et al., 2025) and 2021 in this year's report, are different due to: (i) updated data being published; (ii) by adding new data at the end of the time series for each indicator and removing data at the start of the time series, the highest and lowest scores used for calculating normalised scores across all countries and all years for an indicator can change, directly impacting the normalised scores, and (iii) the rolling 11-year period affects the imputation of missing data, with some countries that in earlier reports were reporting data for an indicator not doing so in later reports, and hence, imputations are used instead.

Data availability: Availability of raw data values for the 506 country-year observations included in the IOI 2025 is shown in **Table 1**. For all indicators except ENT, data are available for at least 92% of the country-year observations, which is above the threshold suggested by the JRC (65%). The relatively lower data availability of the ENT indicator is attributed to the frequency of the Community Innovation Survey (CIS) waves. For example, for the EU27 and other countries participating in the EU CIS, data are available for every other year, in this case for 2014, 2016, 2018, 2020, and 2022.

The latest available year is no less than 2022 for every variable, while raw data are most recent for the KIABI indicator (2024). Thus, the lag between latest year of data across indicators is at most two years, above the threshold suggested by the JRC (five years). Furthermore, the latest available data point in at least 65% of the assessed countries corresponds to the latest available year,

¹⁸ All calculations for the development of the IOI were conducted through the COINr package (Becker *et al.*, 2022).

¹⁹ That is, 46 units (45 countries and the EU27 aggregate) for 11 years (2014 to 2024).

suggesting that in the vast majority of the countries (roughly 90% or more with the exception of ENT indicator) the data used are very recent and close to the year in which data collection was made (2025).

Imputation of missing data: A three-stage procedure is followed for imputing missing data points:

- Missing data in the beginning or the end of the time period 2014 to 2024 are imputed using the last known value. That is, in case data was missing for one country-year observation for 2024, the last known value was imputed from the previous year.
- If data was missing between two non-missing data points, the average of those adjacent years was used.
- In case no last known values are available (i.e., an indicator is not reported for a particular country for all years), the imputation was done using the MICE package in R (Van Buuren and Groothuis-Oudshoorn, 2011; Van Buuren, 2018), which implements multivariate imputation by chained equations. For the IOI 2025, the JRC imputed the KIABI indicator for China, the ENT indicator for Canada, and Mexico, the trademark indicator for Greece, and Ireland, and the GOOD_VA and SERV_VA indicators for North Macedonia, and Serbia using MICE.

All indicator data shown in Section 3.2 and descriptive statistics and correlations in this Annex exclude the MICE imputed data. Please refer to **Table 9** in **Annex 3** for statistics of the data including the MICE imputations. **Figure 18** up to **Figure 23** in **Annex 3** show the available data for each indicator by country and year.

Outlier treatment: No outliers were detected using the criteria set out in JRC/OECD (2008), according to which (absolute) skewness and (absolute) kurtosis are considered excessive when they simultaneously cross the threshold-values of 2 and 3.5, respectively (Groeneveld and Meeden, 1984). Descriptive statistics for the non-normalized IOI 2024 dataset are shown in **Table 2**.

Table 1. Availability of raw data values for the country-year observations in IOI 2025.

Indicator	available data points (number)	available data points (share)	latest available year	countries with non-missing data points corresponding to the latest available year
GOOD_VA	484	95.65	2022	95.65%
SERV_VA	484	95.65	2022	95.65%
KIABI	481	95.06	2024	89.13%
PCT_POP	506	100.00	2022	100.00%
TRA_POP	467	92.29	2023	95.65%
ENT	231	45.65	2022	65.22%

Source: Authors' calculation.

Normalisation: After imputation, the data was normalised over the last eleven years of all countries (country-year observations). The min-max normalisation was applied across all years and ensures the maximum score of 10 corresponds to the maximum observed value over the 11 years while the minimum score, zero, corresponds to the lowest observed value over the 11 years. Descriptive statistics for the normalised indicators are provided in **Table 3**.

Table 2. Descriptive Statistics of IOI indicators for the last year of iteration (2024) [before treatment and normalisation].

Indicator	Min	Max	Mean	Median	Std	Skew	Kurt	N. non-missing	% non-missing
GOOD_VA	2.54	58.9	29.3	26.1	14.5	0.39	-0.73	43	96%
SERV_VA	12.7	69.3	35.3	34.2	10.3	0.91	2.23	43	96%
KIABI	7.3	28.4	15.9	15.7	4.37	0.42	0.47	44	98%
PCT_POP	1.56	404	97.3	56.4	110	1.35	0.96	45	100%
TRA_POP	421	7660	2630	2180	1540	1.27	1.87	43	96%
ENT	8.8	77.4	48.4	52	14.2	-0.38	0.1	42	93%

Source: Authors' calculation.

Weighting: All indicators and components of the IOI are equally weighted within their aggregate. Unequal weights were used in previous editions of the IOI so that the IOI was statistically balanced in its underlying components. After the methodology update implemented in the 2023 edition of the index, this was not necessary.

Aggregation: All indicators were added up using the arithmetic mean to form components. All components of the IOI were added up using the arithmetic mean using equal weights.

Table 3. Descriptive statistics of the normalised IOI indicators for the last year of iteration (2024).

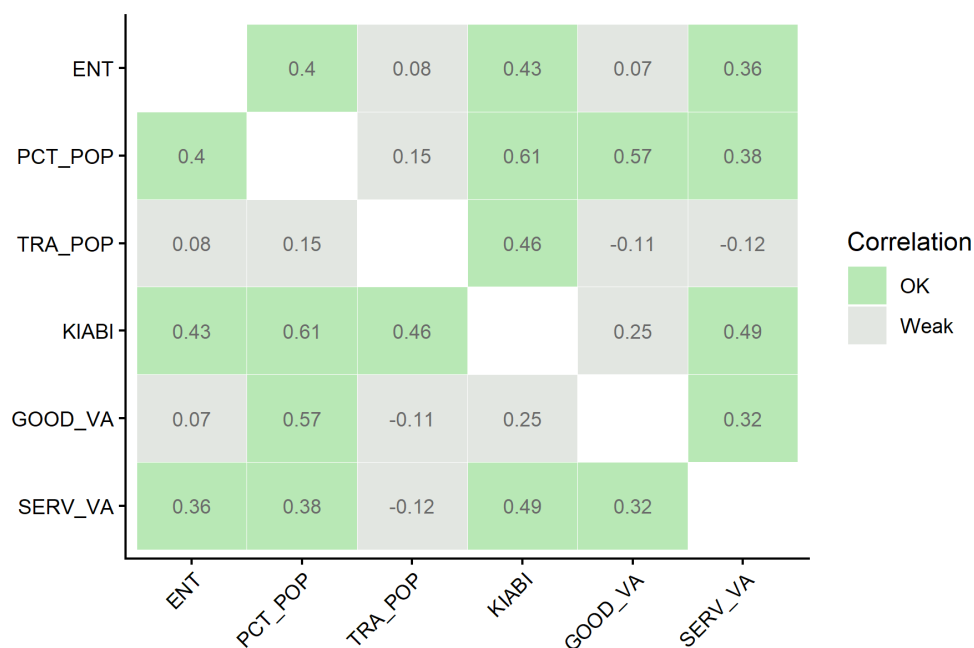
Indicator	Min	Max	Mean	Median	Std	Skew	Kurt	N. non-missing
GOOD_VA	0.00	8.63	4.03	3.56	2.19	0.46	-0.65	45
SERV_VA	0.26	9.71	4.06	3.92	1.68	0.88	2.36	45
KIABI	0.70	10.00	4.50	4.44	1.90	0.43	0.55	45
PCT_POP	0.03	9.92	2.39	1.38	2.69	1.35	0.96	45
TRA_POP	0.04	5.74	1.80	1.48	1.20	1.22	1.82	45
ENT	0.00	9.70	5.61	6.02	1.99	-0.37	0.04	45

Source: Authors' calculation.

Correlations between normalised indicators and the structural coherence of the IOI

The JRC/OECD guidelines on composite indicators (JRC/OECD, 2008) recommend correlations between 0.30 and 0.92 for any two indicators that belong to the same aggregate. An adequate correlation ensures a minimum level of information (in this case at least 9%) transmitted from indicators to aggregates. **Figure 10** shows that the indicators of the intellectual property component (TRA_POP and PCT_POP) do not meet these criteria and have a slightly weaker correlation (0.15). Nevertheless, both indicators are included in the IOI to cover innovations in both goods and services. The correlation between the indicators of the Technology Capacity component (GOOD_VA and SERV_VA) are above the 0.3 threshold (0.32).

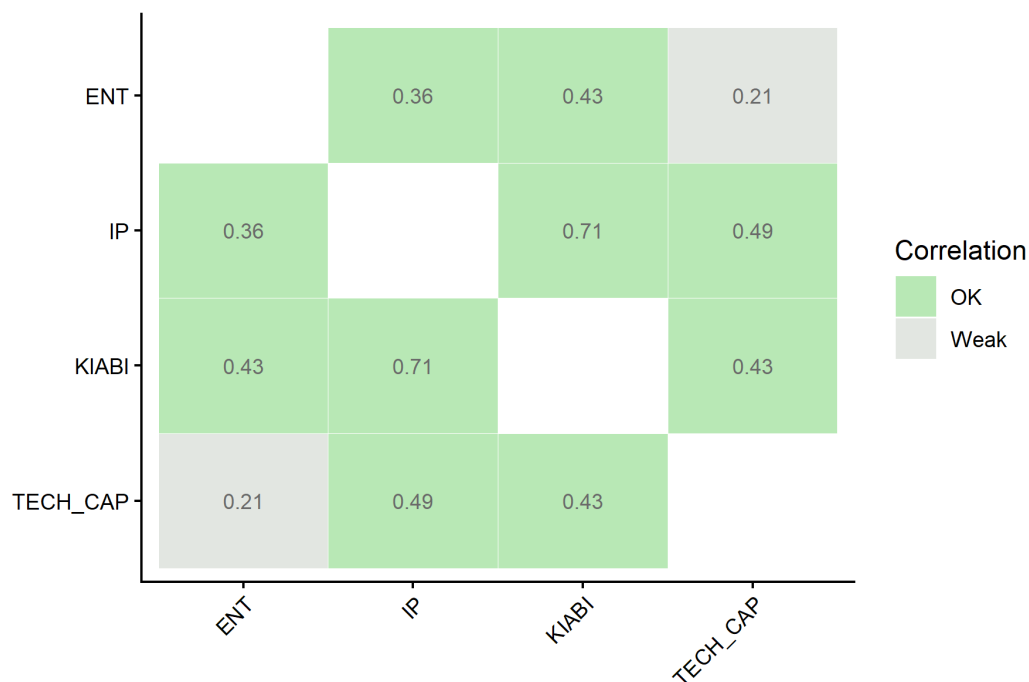
Figure 10. Pairwise Pearson correlations between indicators.



Source: Authors' calculation.

Figure 11 shows correlations between the four components of the IOI. Overall, the correlation structure is adequate. All correlation coefficients except the one between TECH_CAP and ENT are above the threshold of 0.3.

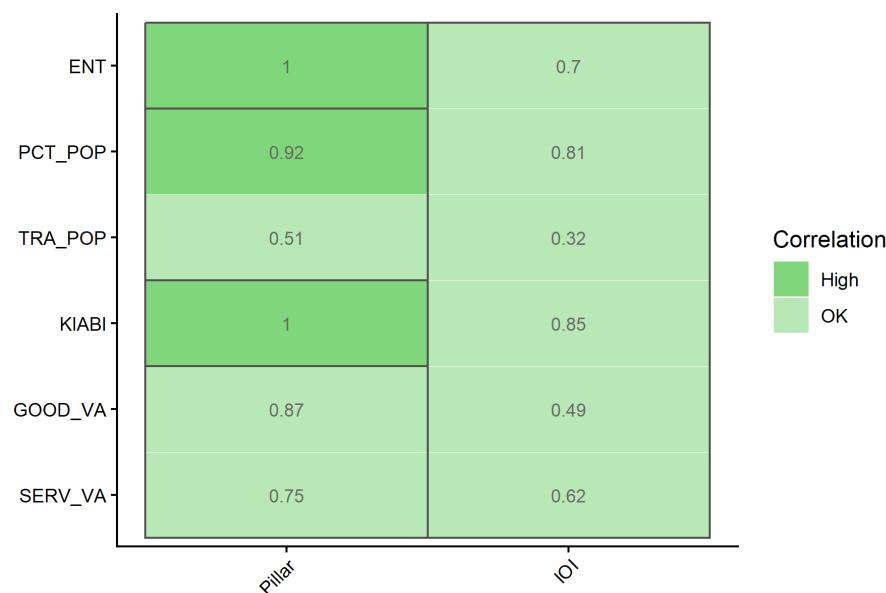
Figure 11. Pairwise correlations between components of the IOI.



Source: Authors' calculation.

Figure 12 depicts the pairwise correlations between indicators with their respective components and the IOI. From the left column, we see that the PCT_POP and TRA_POP indicators are both well correlated with the IP component, and the same holds for the GOOD_VA and SERV_VA indicators and the TECH_CAP component. The right column shows indicator correlations with the IOI. The strength of the correlation between indicator and aggregates reflects the amount of information (that is, discrimination among countries) that an indicator transmits to the index, as it measures the amount of variation in the Index that can be explained by the variation in the indicator. All six correlations of individual indicators with the IOI fall within the recommended guideline, providing evidence that the IOI effectively summarises the information from its six indicators. However, the individual indicators' impact to the overall IOI scores appears to vary. The KIABI and PCT_POP indicators have the strongest correlation with the IOI while TRA_POP has the lowest, followed by GOOD_VA, meaning that variation in PCT_POP is more adequately reflected in the IOI scores and rankings than the variation in TRA_POP. This is evident also from the difference of correlations between the two indicators and IP component (0.92 and 0.51 respectively). In the next iteration of the IOI, JRC will investigate the possibility of balancing the contribution of the two indicators to their component and the IOI by considering unequal weights among them. On the other hand, the correlations of GOOD_VA and SERV_VA with the IOI differ less from one another, suggesting that the two components of the TECH_CAP component have a more balanced contribution to the IOI scores.

Figure 12. Pairwise correlations between indicators and their respective components and with the IOI.

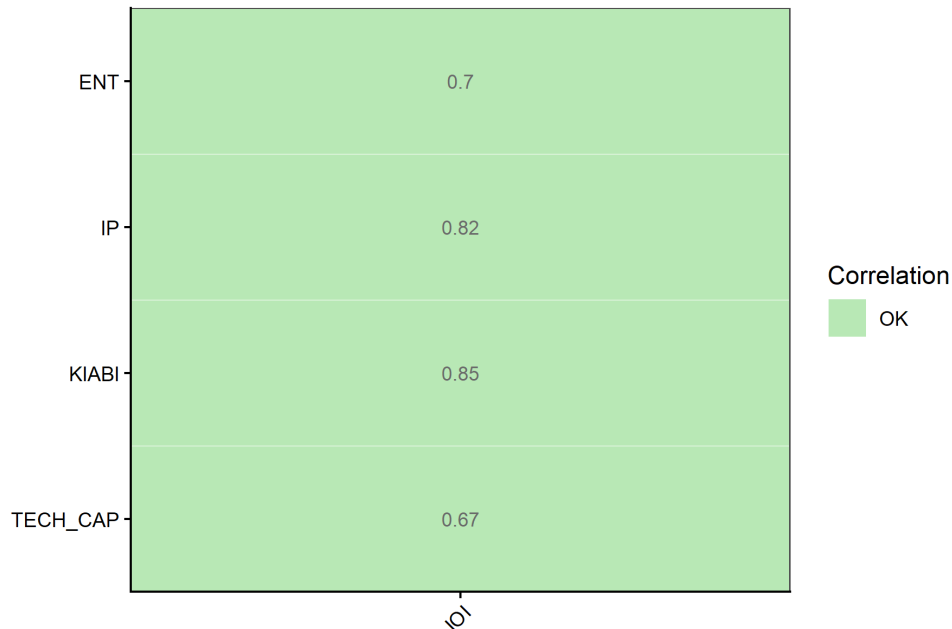


Source: Authors' calculation.

Figure 13 depicts the correlations between the IOI and its components. At this level, which is the most important level of aggregation, we see that all components have an adequate correlation with the IOI. The closeness between the correlations (ranging from 0.67 to 0.85) indicates that each component contributes information to the IOI to a rather similar degree and there is no domination of the IOI by a specific component. In addition, all correlations are very close or above the threshold of 0.7, which suggests that the variation in the Individual components can explain roughly 50% or more of the variation in the IOI. The TECH_CAP and ENT components have a slightly lower correlation, as a result of their low correlation with the other components. Nevertheless, all

correlations are adequate and similar, providing evidence that the IOI is a good summary measure of the information contained in the six considered indicators.

Figure 13. Pairwise correlations between the IOI and its components.



Source: Authors' calculation.

Principal components analysis for the four components

The JRC conducted a PCA analysis to confirm that the four components of the IOI capture a single latent phenomenon. Ther results suggest that there is only one principal component with an eigenvalue greater than one (2.35), which explains 58.8% of the variance in the IOI. Both these values fall within the thresholds suggested by the relevant literature, suggesting that the IOI indeed captures a single latent phenomenon.

Overall, the correlation and principal components analysis suggest that the IOI is a statistically coherent structure, with all its indicators adequately reflected in their components and the final index. The four components contribute adequately and in a balanced way to the IOI, while they do summarize country performance with respect to a single latent phenomenon.

Annex 2. Robustness of IOI ranks and validation of results

Uncertainty analysis

Methodological choices are an unavoidable part of constructing a composite indicator. Developers make these choices using expert opinion, statistical theory or other practical considerations. However, there often exist equally valid alternatives to the choices being made to develop a composite indicator. In this sense, methodological choices introduce some uncertainty, as users may ask whether rankings are robust to these alternative specifications. The ranking of any composite indicator should not be heavily dependent (i.e., there should not be much uncertainty) to a particular methodology chosen during its development. This section investigates to what extent modelling choices influence the ranking of countries in the IOI.

The uncertainty analysis systematically changes four different modelling assumptions: the normalisation, imputation, the weights and aggregation. **Table 4** lists the model specifications used for the nominal IOI and the alternatives considered in the uncertainty analysis. Mean imputation is the alternative method to test for uncertainty in the treatment of missing data. Mean imputation is a frequently used imputation method in composite indicators. It substitutes missing data for an indicator with the mean of the country's remaining indicators in the corresponding aggregation level. The standardization of indicators through z-scores with a mean of five and a standard deviation of 1.5 is the alternative normalisation method examined. Previous versions of the IOI used this normalisation formula. With respect to the aggregation formula, JRC considered two alternative aggregation functions to the arithmetic mean, which are variants of the generalized mean:

Equation 1: The generalised mean

$$M_p(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n x_i^p \quad 1/p$$

in which the exponent p can be used to control for the level of compensability allowed in the aggregation. The arithmetic mean corresponds to a generalized mean with $p=1$, while if $p \rightarrow 0$ the generalized mean converges to the geometric mean. The JRC-COIN considered two alternatives with $p=0.5$ and $p \rightarrow 0$ respectively, each of which implies a limited level compensability compared to the arithmetic mean. In these alternatives, countries showing unbalanced performances are being increasingly penalized as p decreases.²⁰ Lastly, the uncertainty analysis varies the aggregation weights randomly around +/-20% from their nominal values.

Table 4. Uncertainty parameters: missing values, normalisation, aggregation and weights.

Source of Uncertainty	Original	Alternative
I. Treatment of missing values	EM algorithm	Mean imputation
II. Normalisation of indicators	Min-max across pooled sample of 11 years	Z-score (mean = 5, SD = 1.5) across pooled sample of 11 years
III. Aggregation formula at component level	Arithmetic mean (equivalent to a generalised mean in (1) with $p = 1$)	1. Generalised mean with $p = 0.5$ 2. Generalised mean with $p \rightarrow 0$ (geometric mean)
IV. Component weights	Equal weights	+/- 20%

Source: Authors.

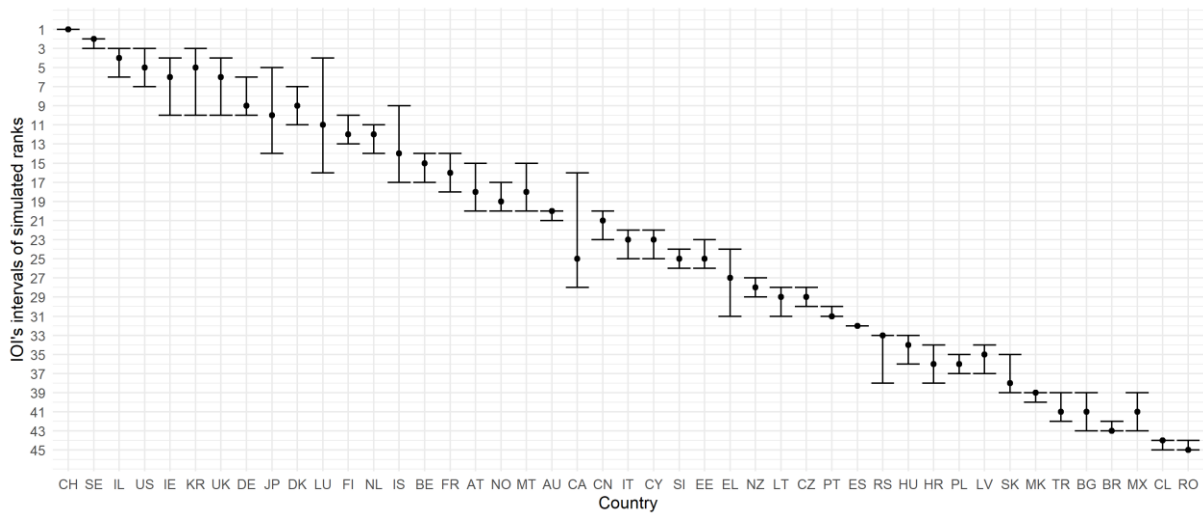
All the combinations (original and alternative) of the normalisation, imputation and aggregation steps result in twelve alternative versions of the IOI. The Monte Carlo uncertainty analysis employed (see Saisana et al., 2005; Salteli et al., 2008) analysis consists of randomly computing one of these

²⁰ To elaborate more on the limited compensability under different p values, assume two components A and B being aggregated into a composite index with equal weights, where $A=0.5 \times B$. If $p=1$, (arithmetic mean), a decline in A (the smaller component) of 1 unit will cause the same decline in the composite index as a decline in B (the larger component) by 1 unit *perfect compensability. If $p=0.5$, the decline in A will cause a decline in the composite index which will be about 1.41 times larger compared to a decline in B by the same amount, while when $p \rightarrow 0$ (geometric mean) the decline in A will cause a decline in the composite index which will be two times larger compared to a decline in B by the same amount (see the discussion on the properties of the generalised mean in UNDP, 1997).

versions 4000 times using a randomly selected set of weights from a distribution of weights with random perturbation of up to 20% of their original value. This results in 4000 alternative versions of the IOI. The statistics of interest - plotted in **Figure 14** and shown in **Table 5**- are the median rank, the 5th and 95th percentile rank for each country across the 4000 versions.

As **Figure 14** shows, the ranks of the IOI are robust against changes in methodology. The dots in **Figure 14** are the median rank across the simulations. These are ordered closely to the original ranking of the IOI as indicated by the country order on the x-axis. In particular, 97.8% (45/46) of countries' median rank is less than three ranks away from the nominal ranking. These are cases like United States, China, and Ireland which do switch ranks, but only by a few places. Notably, for roughly half of the countries, (22/46) the median rank coincides with the nominal rank. There is only one country for which the median rank deviates more than three positions from the nominal rank (Canada, four rank positions). The likely cause for the differences in this case is the existence of missing data, and consequently, to the imputation process, in which the default modelling choice (the EM imputation) takes into account the country's performance in a set of additional indicators highly correlated with the indicator to be imputed. Also, Canada does not have data for ENT which is an indicator but also a component in IOI, and thus its overall weight in the IOI is somewhat larger compared to other indicators such as, e.g., patents and trademarks (0.25 instead of 0.125),

Figure 14. Median rank in the Innovation Output Indicator across simulations with alternative methods.



Source: Authors' calculation.

The confidence interval (CI) provides another indication of the uncertainty in the ranking due to modelling choices. The confidence interval reflects whether modelling alternatives result in a wide variety of rankings. In the uncertainty analysis, confidence intervals are computed as the 5th and 95th percentile of the observed distribution of ranks across simulations. The CI indicates highly stable rankings for 26 out of 46 countries (56.5%), which have a CI range of zero to three. An additional 15 countries have a CI range between four and seven, indicating some uncertainty, but still allowing accurate inference. Lastly, there are only four countries (Canada, Iceland, Japan, and Luxembourg) that have a CI of greater than seven positions, which indicate a larger uncertainty in their ranking and their ranks should be interpreted with some caution. **Table 5** shows the median ranks, confidence intervals, and deviations from the nominal rank.

Table 5. Table of the Median rank, Q5 and Q95 across simulations and nominal rank.

Nominal IOI Rank	Country	Median Rank	Q5	Q95	CI Range	Deviation from Nominal
1	CH	1	1	1	0	0
2	SE	2	2	3	1	0
3	IL	4	3	6	3	-1
4	US	5	3	7	4	-1
5	IE	6	4	10	6	-1
6	KR	5	3	10	7	1
7	UK	6	4	10	6	1
8	DE	9	6	10	4	-1
9	JP	10	5	14	9	-1
10	DK	9	7	11	4	1
11	LU	11	4	16	12	0
12	FI	12	10	13	3	0
13	NL	12	11	14	3	1
14	IS	14	9	17	8	0
15	BE	15	14	17	3	0
16	FR	16	14	18	4	0
17	AT	18	15	20	5	-1
18	NO	19	17	20	3	-1
19	MT	18	15	20	5	1
20	AU	20	20	21	1	0
21	CA	25	16	28	12	-4
22	CN	21	20	23	3	1
23	IT	23	22	25	3	0
24	CY	23	22	25	3	1
25	SI	25	24	26	2	0
26	EE	25	23	26	3	1
27	EL	27	24	31	7	0
28	NZ	28	27	29	2	0
29	LT	29	28	31	3	0
30	CZ	29	28	30	2	1
31	PT	31	30	31	1	0
32	ES	32	32	32	0	0
33	RS	33	33	38	5	0
34	HU	34	33	36	3	0
35	HR	36	34	38	4	-1
36	PL	36	35	37	2	0
37	LV	35	34	37	3	2
38	SK	38	35	39	4	0
39	MK	39	39	40	1	0
40	TR	41	39	42	3	-1
41	BG	41	39	43	4	0
43	MX	41	39	43	4	2
44	CL	44	44	45	1	0
45	RO	45	44	45	1	0

Source: Authors' calculation.

Sensitivity Analysis

Sensitivity analysis builds on the results from an uncertainty analysis to investigate which modelling assumptions drive variation in the rankings of countries. JRC implemented a variance-based sensitivity analysis (see Salteli et al, 2000; Saisana et al., 2005) in which we vary the same factors as in the previous uncertainty analysis (given in **Table 3**). The output of this analysis is the share of variation in the average differences between the nominal ranking and the ranking produced by each of 1000 iterations in which the uncertain factors (normalisation, weights, imputation, and aggregation function) are randomly varied. In our analysis we also allow for interaction between uncertain factors and capture the share of variation attributed to the interaction of each uncertain factor with others. **Table 6** shows the results of this analysis, in which S_i (called the first-order Sobol sensitivity index) provides the share of variation in the target output explained by the variation in each uncertain factor alone, while ST_i (termed the total sensitivity index) is the share of variation in the target output explained by the variation in each uncertain factor alone and its interactions with other uncertain factors.

Table 6. Sensitivity indices from variance-based sensitivity analysis on the average rank deviations of countries in the IOI.

uncertain factor	Normalisation	Weights	Imputation	Aggregation	sum
S_i	32.9%	21.3%	2.5%	29.1%	85.8%
ST_i	54.3%	33.2%	11.4%	58.7%	

Source: Authors' calculation.

As seen from **Table 6**, varying the normalisation and the aggregation functions explains roughly 29-32% of the variance in the average rank differences. Varying weights alone explains roughly 21% of the variance in rank differences, while imputation explains very little share of the variance. All the uncertain factors, taken singly, explain roughly 86% of the variance in the average rank differences. The remaining 16% is the fraction of the variance that is not explained by the input factors taken singly, but by interactions between the factors themselves. Accounting also for interactions among uncertain factors, normalisation and aggregation functions arise as the most influential as they account for more than 50% of the variance in the average rank deviation between nominal ranking and simulated rankings. Note that the high influence of these two options in the rank variation does not suggest that the IOI ranks are sensitive to these choices. Instead, the results in **Table 6** explain the factors responsible for the (satisfactorily low) variation in ranks observed in **Table 5** and **Figure 14**.

Index validation analysis

As a composite indicator, the IOI summarizes a latent concept that cannot be directly measurable by a single metric. Its construction requires the use of metrics of observable phenomena (such as the patent and trademark data analysed in the previous section) that are part of or closely related to its target concept. Therefore, it is important to validate the resulting measurement by examining the concordance of IOI scores with two groups of variables: (i) alternative measures of national innovation performance or comparable constructs for the same countries, and (ii) metrics of observable phenomena that resemble or are closely related to innovation output, such as inputs to innovation or framework conditions known to facilitate the process of innovation. A strong relationship between the IOI and such metrics would provide further evidence that the IOI adequately approximates the innovation output performance of the assessed countries.

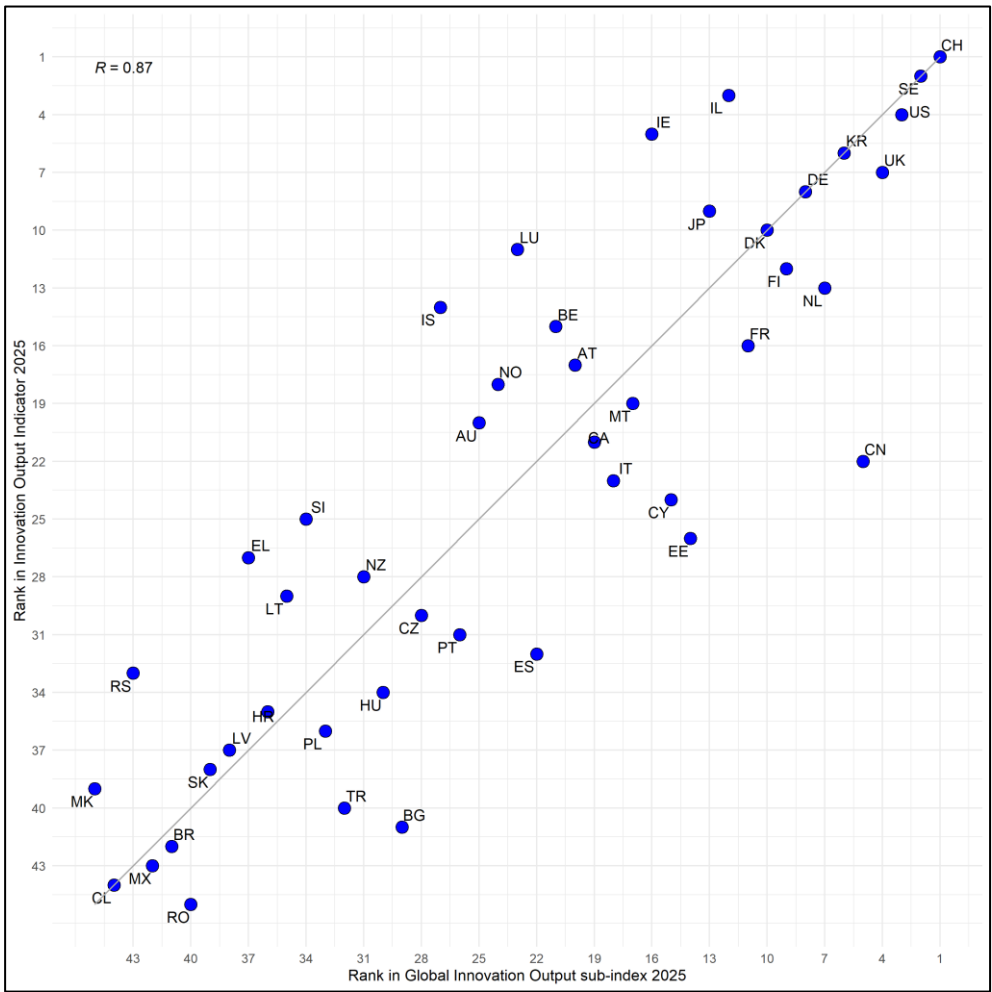
We first compare the findings of the IOI with those of two other summary measures of innovation at the national level, namely the 2025 Summary Index of the EIS (European Commission, 2025a) and its main components, and the Global Innovation Index (GII), developed by WIPO and based on 80 indicators (WIPO, 2025). The GI includes an Input and an Output sub-pillar, of which the latter is conceptually most related to the IOI. We visually depict the relation between these indices, their components and the IOI and measure the strength of their association through correlations.²¹

Figure 15 plots the IOI 2025 ranks against that of the Global Innovation Output Sub-index 2025 for the same sub-sample of countries. Overall, there is a strong positive correlation between the two measures of innovation output, indicating that both measure a similar phenomenon. Certain countries are found consistently at the top or at the bottom of the rankings in both measures (such as, for example, Switzerland which ranks 1st in both, and Mexico which ranks 43rd in the IOI and 42nd in the GI Output Sub-Index among countries included in both). However, there are notable differences in the relative performance of countries across the two measures. This can be seen from the fact that many countries diverge from the “equal rank” line: countries above the line are ranked higher (within the common sub-sample of countries in both indices) in terms of the IOI than in terms of the GI Output Sub-index (and vice versa for those below the line). It is also evident from the differences in country ranks by the two indices, which is on average equal to 5.15 positions²². These differences can be attributed to the wider scope of the GI Output Sub-Index, which also monitors creative outputs such as creative goods and services and online creativity, not covered by the IOI. For instance, Cyprus ranks similarly in terms of the IOI and the “knowledge and technology outputs” dimension of the GI output sub-index (24th and 22nd respectively). However, it is ranked much better (11th) in terms of the “Creative outputs” dimension of the GI Output sub-Index. Therefore, the overall rank of Cyprus is much higher in the GI Output sub-index (15th) compared to its IOI rank (nine rank positions). On the other hand, Slovenia is ranked 25th in both the IOI and the “knowledge and technology outputs” dimension of the GI output sub-index. However, it performs relatively worse in terms of the GI “Creative outputs” dimension (41st among the countries included in both indices), resulting in its overall rank in the GI Output sub-index (34th) being nine positions lower than its IOI rank. The country with the largest drop in its IOI rank compared to the GI Output Sub-Index is China (22nd versus 5th). This is attributed to China’s exceptional performance in terms of both Pillars in the GI Output Sub-Index (1st among the country subsample in terms of “knowledge and technology outputs” and 14th in the “creative outputs”). Apart from “knowledge creation” (GI Pillar 6.1), China ranks at the top-10 in terms of “knowledge impact”, “knowledge diffusion” (Pillars 6.2 and 6.3), as well as in “intangible assets” (Pillar 7.1), which includes, estimates of global brand value and industrial designs, in addition to trademarks. On the other hand, Iceland exhibits the largest positive difference between its IOI rank (14th) and its GI Output Sub-Index rank (27th). Iceland performs rather poorly on several of the additional concepts included in the latter (“knowledge impact”, “knowledge diffusion”, global brand value and industrial designs).

²¹ Each of the Figures below shows the subsample of countries that appear in both indexes. In the top left of each Figure, readers can find the Pearson correlation coefficient between the two indices

²² Refers to the average of the absolute differences in ranks as suggested in Saisana *et al.* (2005).

Figure 15. Ranks of the IOI and the Global Innovation Output sub-index



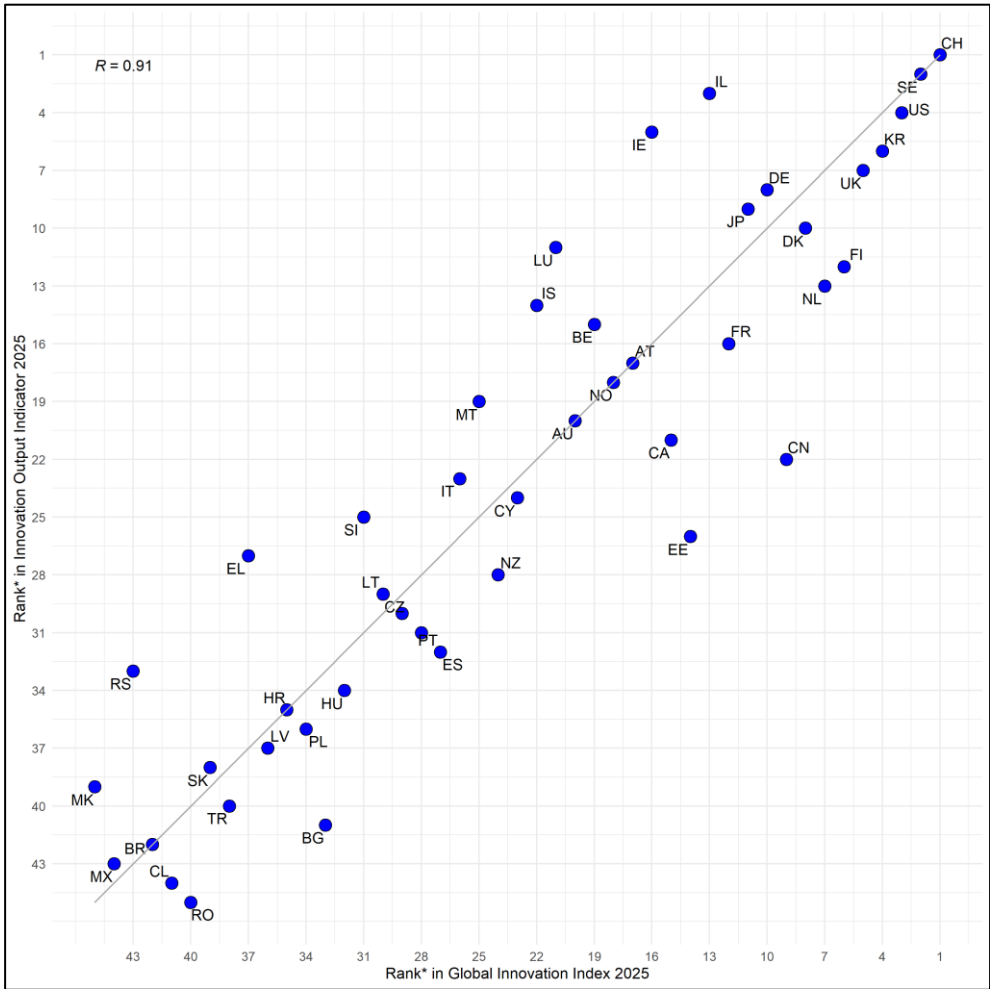
Source: Authors' calculation and WIPO (2025). Notes: Calculation based on the subsample of countries that appear in both indexes. In the top left of each Figure, readers can find the Pearson correlation coefficient between the two indices. The straight line suggests equal rank in the two indices. Countries above the line are relatively better ranked in terms of the IOI than they are in terms of the GII Output Sub-Index and vice versa for countries below the line.

Figure 16 plots the IOI 2025 ranks against those of the Global Innovation Index for the 2025. The correlation coefficient shows a strong relationship between the two indexes, which is slightly stronger than that of the IOI with the GII output sub-index (Pearson correlation coefficient 0.91 vs. 0.87). This can, to some extent, be attributed to the concordance between the indicators considered in the GII Input Sub-Index and the indicators of the IOI. For example, GII Input Sub-Index includes indicators such as tertiary enrolment, graduates in science and engineering, gross expenditure on R&D, and GERD performed by business, that measure concepts which convey strong association with the IOI (see the discussion below and **Table 7**).²³ China is again the country with the largest drop in its IOI rank compared to the GII Output Sub-Index is China (22nd versus 9th). Beyond China's good performance aspects includes in the GII Output Subindex but not in the IOI, as discussed

²³ The Pearson correlation coefficients between the IOI and the five Pillars of the GII Input Sub-Index range from 0.538 to 0.895

above, Cina also performs very well in the Infrastructure (GII Pillar 3, 6th) and Business sophistication (GII Pillar 5, 7th), which includes aspects such as linkages to innovation and knowledge absorption that are not covered by the IOI.

Figure 16. Ranks of the IOI and the Global Innovation Index



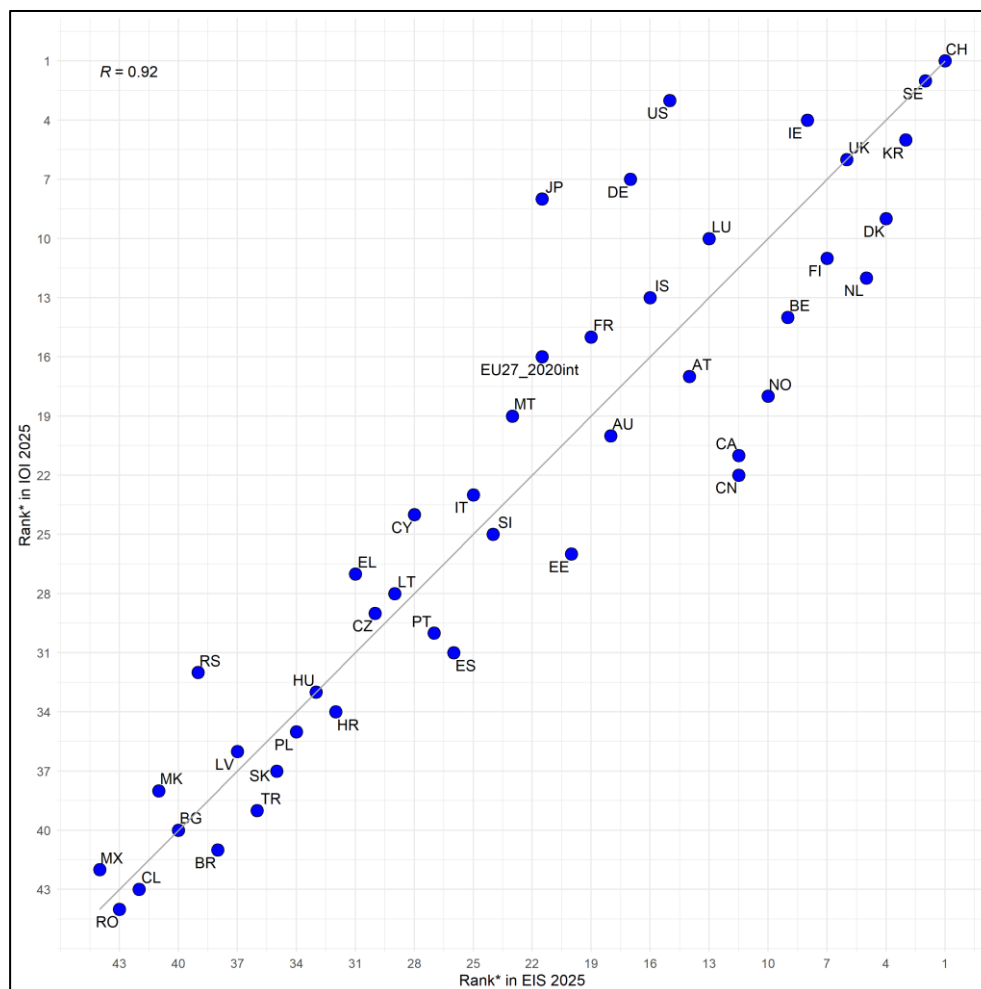
Source: Authors’ calculation and WIPO (2025). Note: Calculation based on the subsample of countries that appear in both indexes. In the top left of each Figure, readers can find the Pearson correlation coefficient between the two indices. The straight line suggests equal rank in the two indices. Countries above the line are relatively better ranked in terms of the IOI than they are in terms of the GII and vice versa for countries below the line.

In **Figure 17**, the IOI 2025 ranks are plotted against those of the European Commission’s (2025a) Summary Innovation Index (SII) for 2025. The Pearson correlation coefficient between the two Indices is large, indicating that the two indices measure a similar underlying phenomenon. This verifies IOI as a suitable complement to the EIS.

It is also informative to consider for comparison the four pillars of the European Innovation Scoreboard (Investment, Framework Conditions, Innovation activities and Impacts). The Innovation activities and Impacts pillars conceptually resonate more with the IOI and each include three dimensions (“Innovators, Linkages, and Intellectual assets”, “Employment and sales impacts, Trade impacts, and Resource and labour productivity” respectively). As expected, the IOI correlates stronger with the Innovation activities and the Impacts pillars, while it also shows a strong correlation with the other two pillars of EIS (see the first column of **Table 7**). In the case of the

Innovation activities pillar, the positive relationship is mainly driven by the high correlation between the IOI and the Linkages and Intellectual assets sub-pillars. These include indicators related to SME innovative activities and patent and trademark applications that are also included in the IP component of the IOI (albeit denominated with population and not GDP as is the case in the EIS). In the case of the Impacts sub-pillar, the strong relationship appears to be mainly driven by the Sales and Employment Impacts sub-pillar which includes the indicator “Employment in innovative enterprises” that is closely related to the ENT indicator and corresponds to the share of innovative enterprises, measured in terms of employment, and the Resource and Labour productivity sub-pillar.

Figure 17. Ranks of the IOI and EIS Summary Innovation Index



Source: Authors' calculation and European Commission (2025). Note: Rank calculation based on the subsample of countries that appear in both indexes. In the top left of each Figure, readers can find the Pearson correlation coefficient between the two indices. The straight line suggests equal rank in the two indices. Countries above the line are relatively better ranked in terms of the IOI than they are in terms of the EIS Summary index and vice versa for countries below the line.

In some cases, a high statistical association among composite indices monitoring the same or similar concepts may indicate redundancy. This can be investigated by assessing whether each composite, which is built on different conceptual framework and choice of indicators than the other composites, manages to provide a distinct perspective on country performance relative to the

others²⁴. As is shown in **Table 7**, this is the case for the IOI, and GII, SII and their components. The third and fourth columns report the average rank deviation between these indices and their components in comparison with the IOI, as well as percentage of countries where the IOI ranking diverges by five positions or more from the rankings of other indices or components. While the correlation coefficients suggest strong alignment, the rank differences between the IOI and the two indices and their components are significant.

In the case of the GII and its two sub-indices, these are on average equal to a difference of about 4 to 5 rank positions, which is roughly 10% of the sample of countries included in both indices (45), while in at least 38% of the countries the IOI rank deviates more than 3 positions from the corresponding GII or sub-index rank. This is reflected, for example, in the rank of Luxembourg (11th in the IOI but 21st in terms of the GII) and in that of China (22nd in the IOI but 9th in terms of the GII). In a similar fashion, average rank deviations between the IOI and SII or its sub-components range from 4.2 (SII) to 10.4 positions (Trade impacts), while at least 38% of the countries exhibit rank deviations of 5 positions or more.

Table 7. Correlations and rank deviations between GII, EIS, their components, and the IOI

	correlation with IOI	Average rank deviation with IOI	Countries with rank deviations of five or more positions with the IOI (%)
Global Innovation Index	0.905	4.044	38%
• GII Input Sub-Index	0.895	5.022	47%
• GII Output Sub-Index	0.853	5.156	51%
EIS Summary Innovation Index ¹	0.929	4.205	38%
• Investments	0.861	7.543	51%
• Framework Conditions	0.838	7.429	53%
• Innovation Activities	0.874	7.714	58%
– Innovators	0.585	9.771	53%
– Linkages	0.837	7.486	53%
– Intellectual assets	0.631	9.429	58%
• Impacts	0.871	6.971	49%
– Sales and employment Impacts	0.612	10.114	60%
– Trade Impacts	0.423	10.400	53%
– Resource and labor productivity	0.810	7.943	53%

¹ The EIS Summary Index is calculated by averaging directly the 32 indicators of its framework. For the purposes of the analysis in this report, EIS pillar and sub-pillar values were calculated by the authors by aggregating hierarchically the component indicators.

Source: Authors' calculation, WIPO (2025) and European Commission (2025).

We next consider the association between the IOI and its four components with a set of variables that commonly used in the literature to represent inputs to innovation or to assess the quality of

²⁴ Differences in the rankings of two composite indicators monitoring similar concepts (in this case innovation performance) are expected due to differences in the conceptual framework, scope, and the choice of indicators. If despite these differences, rankings to not differ in a meaningful way, then this would suggest redundancy.

innovation-enabling framework conditions within a country.²⁵ For example, R&D expenditure as a share of GDP is widely regarded as a vital input to innovation (R&D expenditure is also part of the EIS “Investments” Pillar), while higher GDP per capita is viewed as a critical enabler of innovation activities. In parallel, a workforce characterized by a higher share of higher education graduates, individuals participating in lifelong learning, and adequate digital skills is considered essential for fostering innovation. These human capital factors, combined with high-speed internet access, cloud computing adoption, and a high share of ICT specialists, collectively reflect the quality of a country’s digital infrastructure and the degree of digitalization within its enterprises. These elements form a complementary ecosystem that underpins the conditions for innovation to thrive.

Table 8. Pearson correlation coefficient between the IOI and its components and contextual indicators

	IOI Components				
	IOI	IP	TECH_CAP	KIABI	ENT
GDP per capita, current prices	0.658¹	0.503	0.216	0.776	0.452
Population (25-34) having completed tertiary education (%)	0.615	0.541	0.197	0.691	0.363
Population (25-64) participating in lifelong learning (%)	0.693	0.724	0.301	0.564	0.514
Individuals (16-74) who have above basic overall digital skills (%)	0.709	0.603	0.368	0.615	0.544
New doctorate graduates in STEM per 1000 population aged 25-34	0.764	0.580	0.494	0.686	0.554
Foreign doctorate students (% of all doctorate students)	0.709	0.729	0.313	0.772	0.347
International scientific co-publications per million population	0.686	0.517	0.173	0.714	0.595
GERD (% of GDP)	0.732	0.661	0.677	0.446	0.514
BERD (% of GDP)	0.690	0.619	0.688	0.442	0.455
HERD (% of GDP)	0.551	0.473	0.267	0.252	0.591
GOVERD (% of GDP)	0.093	0.172	-0.012	0.002	0.141
Venture capital expenditures (% of GDP)	0.563	0.437	0.376	0.443	0.445
Direct government funding and government tax support for business R&D (% of GDP)	0.428	0.312	0.335	0.249	0.385
Cloud computing in enterprises	0.682	0.536	0.399	0.613	0.505
ICT specialists (% of total employment)	0.668	0.509	0.315	0.775	0.395

¹ Correlations based on pairwise complete observations. Correlations larger than 0.5 are highlighted in bold. All correlations except those in grey colour are statistically significant.

Source: All Eurostat (*rd_e_gerd*, *inn_cis* datasets), world Bank (*NY.GDP.PCAP.PP.CD*), OECD MSTI database, European Commission (2025a), & authors’ calculation.

Table 8 shows that the correlations between the IOI and most of the selected variables are very strong (above 0.5) and statistically significant, and the same holds to a large extent for the IOI components. Particularly, the IOI, and its IP, KIABI, and ENT components are all highly (above 0.5) or adequately (above 0.3) correlated with GDP per capita and the set of variables capturing workforce characteristics facilitating innovation such as the share of population aged 25-34 with a tertiary education degree, individuals with adequate digital skills, STEM doctoral graduates, and international scientific co-publications per million population, capturing the quality of scientific research within a country (European Commission, 2025b). Also, the TECH_CAP component is

²⁵ Variable selection was informed from the “Input” and “Framework Conditions” Pillars of the European Innovation Scoreboard and similar Pillars in other composite indices measuring concepts similar to innovation such as, for example, competitiveness.

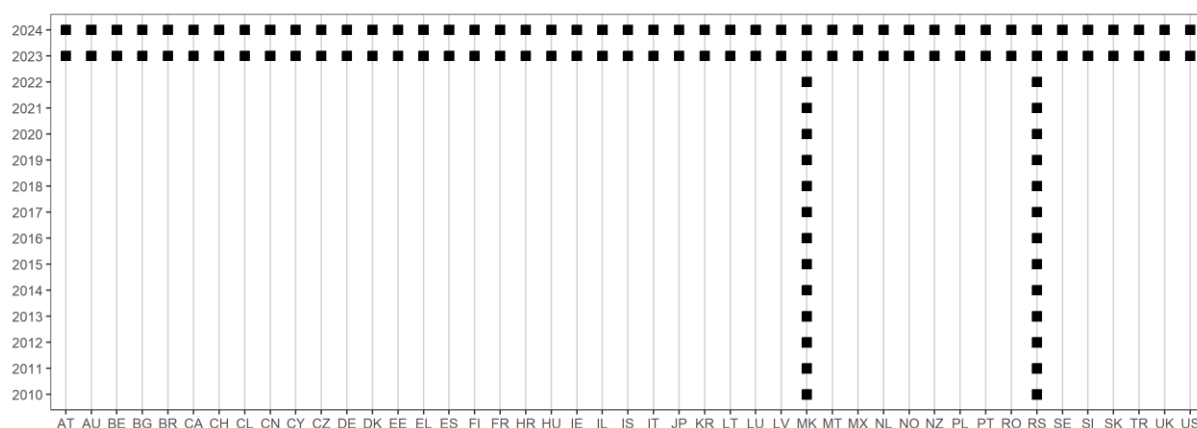
adequately correlated with the variables related to lifelong learning, digital skills, STEM doctorates and foreign doctoral students, suggesting that a diverse and highly, continuously, and digitally educated workforce relates significantly to a country's lower reliance on foreign inputs to produce its knowledge-intensive manufacturing and service exports.

In a similar fashion, the IOI and most components are strongly correlated to indicators capturing R&D and venture capital expenditures and government support to R&D. The high correlations with the total R&D expenditures as a share of GDP (GERD) appear to be driven by the business (BERD) and higher education (HERD) components, while R&D in the government sector is not significantly related to the IOI and its components. This can mask a lag effect, as the correlation captures the association between the latest available data used for both variables, and government R&D typically generates indirect, long-term impacts that are less immediately reflected in a country's innovation output.²⁶ Lastly, high-performing countries in the IOI and its components are consistently associated with advanced digitalization in enterprises, as evidenced by cloud computing adoption and a high share of ICT specialists in their workforce, underscoring the role of digital infrastructure in driving innovation output.

The findings of this sub-section can be summarized in three observations. **First**, country performance in the IOI aligns closely with indicators of innovation inputs and enabling framework conditions. **Second**, the IOI exhibits meaningful correlations with other frameworks that assess innovation performance, including those focused on outputs. **Third**, the divergence in rankings between the IOI and the GII, SII, and their components highlights the IOI's unique contribution to innovation monitoring. These results affirm the conceptual validity of the IOI: it consistently measures country-level innovation output performance through distinct metrics and analytical approaches, while also capturing dimensions of innovation that other indices may overlook.

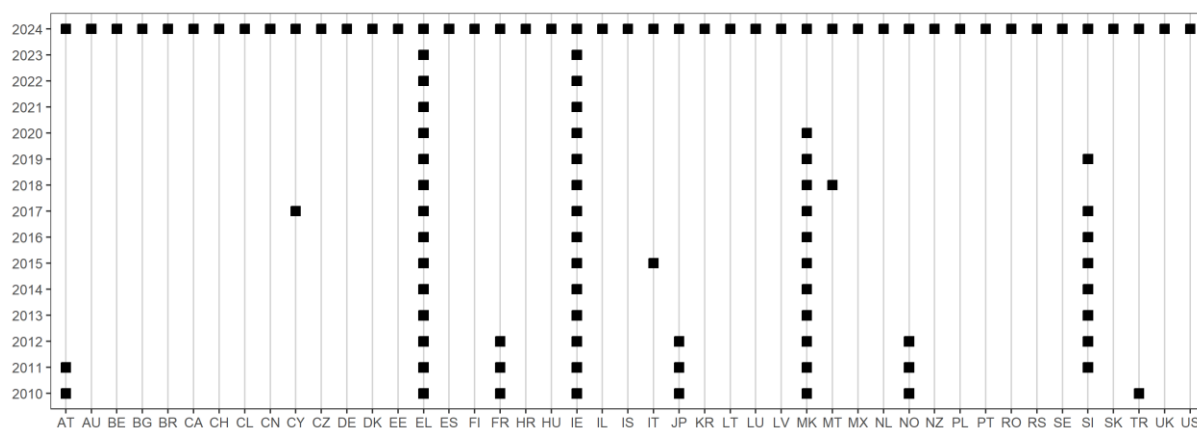
²⁶ Guellec and Van Pottelsbergue De La Potterie (2003) argue that the weak appropriability of basic research results produced in government laboratories and higher education institutions makes it difficult for firms to reap its rewards, while their study finds a negative effect of government intramural R&D on business R&D in 17 OECD countries.

Figure 21. Simplified Missing data years per country on GOOD_VA.



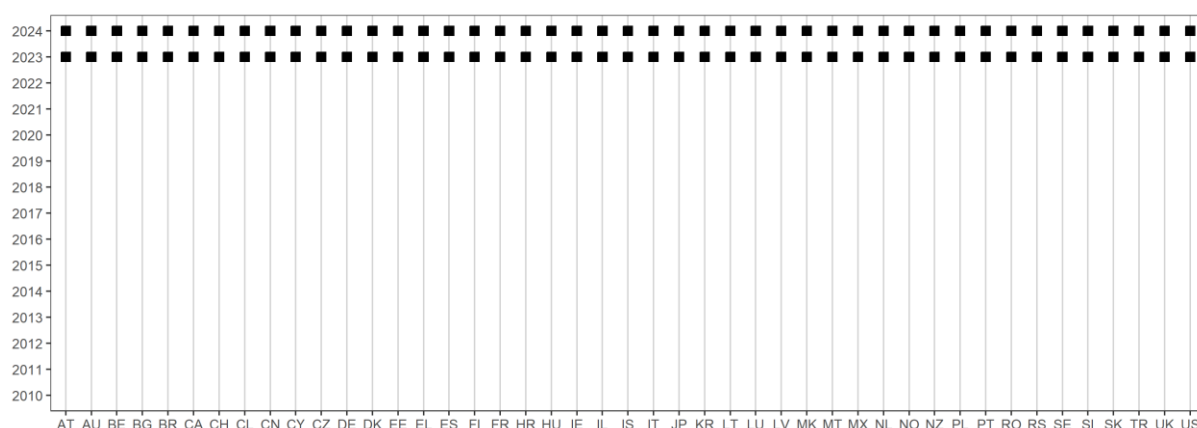
Source: Authors' calculation. Note: Squares indicate missing data for the country-year.

Figure 22. Simplified Missing data years per country on TRA_POP.



Source: Authors' calculation. Note: Squares indicate missing data for the country-year.

Figure 23. Simplified Missing data years per country on PCT_POP.



Source: Authors' calculation. Note: Squares indicate missing data for the country-year.

Table 9. Descriptive statistics of the normalised and imputed IOI indicators for the last year of iteration (2024).

Indicator	Min	Max	Mean	Median	Std	Skew	Kurt	N. non-missing
GOOD_VA	0.06	9.83	4.08	3.78	2.37	0.41	-0.46	46
SERV_VA	0.03	10.00	4.27	4.01	1.99	0.85	1.41	46
KIABI	0.80	10.00	4.61	4.42	1.97	0.39	0.04	46
PCT_POP	0.00	10.00	2.35	1.20	2.74	1.34	0.88	46
TRA_POP	0.09	7.05	1.80	1.38	1.42	1.64	3.33	46
ENT	0.07	9.86	5.87	6.42	2.23	-0.32	-0.43	46

Source: Authors' calculation.

Annex 4. Country performance per IOI indicator

Table 10. Country performance per IOI indicator and change between 2014 and 2024.

Country	Indicator	IOI	GOOD_va	SERV_va	Patents	Trademarks	TECH_CAP	ENT	IP	KIABI
AT	Rank 2024	18	19	20	10	19	17	14.5	11	17.5
	Value 2024	109	4.22	4.22	4.25	1.8	4.22	6.77	3.02	4.76
	% Change	5.83	-5.32	6.53	3.77	-18.52	0.25	5.04	-4.04	20
AU	Rank 2024	21	37	33	22	15	37	3	24	19
	Value 2024	103	2.3	3.35	1.46	2.22	2.83	8.29	1.84	4.68
	% Change	1.98	11.61	1.67	-21.85	7.52	5.49	6.06	-6.44	-2.86
BE	Rank 2024	15	24	10	14	36	16	2	23	14
	Value 2024	116	3.6	5.04	2.84	1.03	4.32	8.69	1.93	5.02
	% Change	16	5.46	0.39	6.88	-19.62	2.44	30.57	-1.71	17.53
BG	Rank 2024	43	39	34	42	25	38	46	37	39.5
	Value 2024	52	2.23	3.09	0.18	1.48	2.66	2.44	0.83	3.08
	% Change	20.93	-0.48	20.25	-4	19.46	10.58	-6.99	16.4	89.19
BR	Rank 2024	44	31	42	45	33	40	37	41	43
	Value 2024	51	2.87	2.44	0.06	1.15	2.65	3.8	0.61	1.74
	% Change	-38.55	-30.17	-19.07	-24.09	399.18	-25.48	-58.13	284.95	15.46
CA	Rank 2024	22	21	32	21	41	25	6	30	31
	Value 2024	98	3.86	3.58	2.01	0.76	3.72	7.89	1.39	3.82
	% Change	-11.71	-13.66	-10.9	-8.33	-48.43	-12.36	-12.03	-24.44	-4.94
CH	Rank 2024	1	2	12	3	5	4	13	2	4
	Value 2024	156	8.4	4.99	8.67	3.26	6.7	6.82	5.97	7.31
	% Change	-8.24	2.18	-13.05	7.6	4.11	-4.08	-31.76	6.62	9.93
CL	Rank 2024	46.5	42	46	41	22	45	45	36	44
	Value 2024	40	2	1.73	0.2	1.57	1.87	2.45	0.88	1.62
	% Change	-6.98	-7.36	-37.76	20.81	56.21	-24.45	16.68	51.28	-25.02
CN	Rank 2024	23.5	9	28	26	4	12	31	17	24
	Value 2024	96	6.69	3.74	1.18	3.61	5.22	4.52	2.39	4.44
	% Change	21.52	9.02	-9.15	246.75	410.45	1.72	-1.54	357.36	36.41
CY	Rank 2024	26	40	18	34	11	29	35	26	8
	Value 2024	92	2.15	4.49	0.48	2.63	3.32	4.37	1.55	6.52
	% Change	0	-34	-41.66	167.05	175.51	-39.38	-7.21	174.18	28.7
CZ	Rank 2024	31.5	17	29	32	27	19	29	33	33
	Value 2024	80	4.56	3.73	0.59	1.34	4.14	5	0.96	3.61
	% Change	3.9	-4.06	29.47	16.98	-39.26	8.6	0.85	-28.81	17.14
DE	Rank 2024	8	5	16	9	13	8	8	7	25
	Value 2024	127	7.34	4.7	5.02	2.33	6.02	7.71	3.67	4.36
	% Change	-0.78	-1.4	1.98	-8.31	13.97	-0.11	-6.02	-2.26	11.24
DK	Rank 2024	9.5	11	4	5	34	7	14.5	6	15
	Value 2024	126	6.03	6.32	6.41	1.05	6.17	6.77	3.73	4.93

Country	Indicator	IOI	GOOD_va	SERV_va	Patents	Trademarks	TECH_CAP	ENT	IP	KIABI
	% Change	15.6	9.52	33.1	26.83	-20.68	20.45	13.24	16.96	12
EE	Rank 2024	28	43	21	33	14	36	20	31	21.5
	Value 2024	89	1.74	4.14	0.49	2.26	2.94	6.27	1.37	4.63
	% Change	32.84	-16.16	28.74	51.14	67.09	11.13	14.43	64.03	84.21
EL	Rank 2024	29	44	14	37	21	33	4	34	38
	Value 2024	88	1.35	4.76	0.3	1.62	3.05	8.01	0.96	3.13
	% Change	12.82	-22.01	-2.86	19.7	-8.16	-7.86	30.34	-4.68	9.23
ES	Rank 2024	34	18	25	28	32	18	38	32	35
	Value 2024	72	4.47	3.92	0.87	1.26	4.19	3.72	1.06	3.48
	% Change	7.46	-2.56	15.06	-5.04	7.48	4.95	6.05	2	19.7
EU27 ¹	Rank 2024	17	6	7	17	20	6	23.5	18	26
	Value 2024	110	7.34	5.57	2.52	1.74	6.45	6.02	2.13	4.23
	% Change	10	-3.01	20.42	-0.92	5.06	5.88	6.77	1.44	26.32
FI	Rank 2024	12	23	5	6	35	14	10	9	13
	Value 2024	121	3.76	5.82	5.77	1.04	4.79	7.45	3.41	5.2
	% Change	4.31	-11.09	14.68	-22.87	-40.16	2.97	20.32	-26.12	16.83
FR	Rank 2024	16	13	30	16	8	15	12	13	21.5
	Value 2024	111	5.76	3.71	2.67	2.91	4.74	6.87	2.79	4.63
	% Change	8.82	-5.45	25.53	-7.82	-13.78	4.68	8.97	-11.03	26.51
HR	Rank 2024	37.5	36	39	36	43	43	32	44	32
	Value 2024	64	2.31	2.68	0.31	0.55	2.49	4.49	0.43	3.66
	% Change	8.47	-26.13	-24.22	18.45	-4.8	-25.12	9.28	2.39	56.6
HU	Rank 2024	36	26	26	29	44	26	44	42	30
	Value 2024	66	3.44	3.85	0.7	0.51	3.64	3.02	0.61	4.05
	% Change	10	-14.35	28.12	19.74	14.67	3.85	-9.7	17.54	39.39
IE	Rank 2024	4.5	3	11	18	9	5	23.5	15	2
	Value 2024	135	8.11	5.01	2.36	2.85	6.56	6.02	2.61	7.93
	% Change	12.5	61.02	9.03	19.08	85.68	36.22	-14.63	48.15	13.21
IL	Rank 2024	3	7	1	7	47	1	25	14	7
	Value 2024	136	6.78	9.71	5.4	0.04	8.25	5.69	2.72	6.66
	% Change	7.09	-2.25	28.83	-13.57	23.33	13.94	0	-13.39	16.32
IS	Rank 2024	14	47	9	15	7	42	1	12	12
	Value 2024	119	0	5.14	2.77	2.98	2.57	9.7	2.88	5.24
	% Change	10.19	-100	-15.48	6.74	-0.86	-17.39	36.16	2.66	-4.8
IT	Rank 2024	23.5	16	45	23	26	31	9	29	28
	Value 2024	96	4.67	1.76	1.45	1.42	3.21	7.67	1.43	4.1
	% Change	11.63	-9.18	-2.26	7.09	30.17	-7.39	14.8	17.4	17.72
JP	Rank 2024	9.5	1	44	2	29	10	18	3	16
	Value 2024	126	8.63	2.02	8.81	1.33	5.33	6.48	5.07	4.84
	% Change	9.57	-12	75.99	5.05	94.31	-2.78	17.41	11.78	15.6
KR	Rank 2024	6	8	27	1	3	11	42	1	3
	Value 2024	134	6.72	3.81	9.92	3.74	5.27	3.64	6.83	7.34
	% Change	20.72	7.67	36.47	78.75	71.96	16.58	15.74	76.84	-2.23
LT	Rank 2024	31.5	38	38	40	23	44	22	35	27
	Value 2024	80	2.24	2.74	0.26	1.55	2.49	6.19	0.91	4.19
	% Change	100	34.27	83.24	-28.29	105.96	57.4	81.74	61.83	206.45
LU	Rank 2024	11	46	37	12	2	46	26	5	1
	Value 2024	125	0.79	2.84	3.23	4.81	1.81	5.65	4.02	10
	% Change	-18.83	-39.73	-27.09	2.11	-47.12	-30.29	-30.19	-34.43	4.13
LV	Rank 2024	39	41	17	30	37	30	43	39	36
	Value 2024	63	2.01	4.49	0.62	0.9	3.25	3.43	0.76	3.44
	% Change	31.25	4.46	54.53	59.75	33.91	34.59	12.5	43.42	50
MK	Rank 2024	41	32	13	46	45	24	36	46	42
	Value 2024	59	2.82	4.82	0.06	0.38	3.82	4.04	0.22	2.07
	% Change	34.09	-3.37	45.88	#NUM!	-38.57	22.77	5.15	-28.81	683.53
MT	Rank 2024	20	27	15	27	1	21	33	10	10
	Value 2024	106	3.38	4.73	0.91	5.74	4.05	4.44	3.32	6.34
	% Change	4.95	28.16	-10.19	104.96	90.7	2.61	-25.77	92.53	11.63
	Rank 2024	45	12	47	47	42	32	40	45	45

Country	Indicator	IOI	GOOD_va	SERV_va	Patents	Trademarks	TECH_CAP	ENT	IP	KIABI
MX	Value 2024	50	5.89	0.26	0.03	0.63	3.08	3.65	0.33	1.52
	% Change	28.21	0.18	#NUM!	-27.4	188.69	4.67	62.81	152.63	3.38
NL	Rank 2024	13	20	24	8	16	22	16	8	9
	Value 2024	120	4.04	3.92	5.18	1.95	3.98	6.71	3.56	6.39
	% Change	7.14	-17.12	-9.83	0.87	-0.78	-13.68	11.5	0.41	25
NO	Rank 2024	19	33	6	13	38	20	7	21	17.5
	Value 2024	108	2.54	5.73	3.14	0.85	4.13	7.74	2	4.76
	% Change	10.2	-32.73	0.32	-9.99	-32.31	-12.83	52.65	-15.92	1.89
NZ	Rank 2024	30	45	40	24	10	47	21	20	29
	Value 2024	82	0.9	2.53	1.4	2.77	1.72	6.23	2.08	4.07
	% Change	-2.38	-17.4	-33.98	-25.4	22.15	-30.32	9.27	0.63	-3.4
PL	Rank 2024	37.5	28	31	39	40	27	41	43	37
	Value 2024	64	3.32	3.59	0.28	0.77	3.46	3.65	0.52	3.3
	% Change	45.45	-14.25	39.52	18.17	17.74	7.23	81.69	17.85	78.57
PT	Rank 2024	33	34	36	31	12	39	28	27	34
	Value 2024	76	2.42	2.9	0.6	2.41	2.66	5.45	1.5	3.52
	% Change	7.04	-10.67	3.7	80.31	50.47	-3.37	-15.72	55.6	70.21
RO	Rank 2024	46.5	15	22	44	28	13	47	40	46
	Value 2024	40	5.5	4.12	0.08	1.33	4.81	0	0.71	1.28
	% Change	-6.98	-13.53	27.75	12.02	103.41	0.37	-100	94.36	141.67
RS	Rank 2024	35	35	19	43	46	28	30	47	39.5
	Value 2024	67	2.41	4.26	0.15	0.22	3.33	4.89	0.19	3.08
	% Change	6.35	0.86	-14.52	69.49	34.5	-9.54	-10.59	46.87	105.88
SE	Rank 2024	2	14	8	4	30	9	11	4	5
	Value 2024	141	5.61	5.34	7.88	1.32	5.47	6.89	4.6	7.18
	% Change	8.46	-6.68	9.22	-2.72	-16.59	0.45	3.61	-4.98	33.61
SI	Rank 2024	27	25	41	25	24	35	17	28	23
	Value 2024	90	3.56	2.52	1.38	1.54	3.04	6.58	1.46	4.45
	% Change	12.5	-18.5	31.44	-7.75	-14.74	-3.27	23.61	-11.57	21.69
SK	Rank 2024	40	29	35	38	31	34	39	38	41
	Value 2024	61	3.04	3.04	0.29	1.32	3.04	3.66	0.8	3.04
	% Change	17.31	-11.83	40.18	43.51	9.93	8.26	2.78	14.74	64.29
TR	Rank 2024	42	30	43	35	6	41	34	25	47
	Value 2024	55	3	2.15	0.45	3.09	2.57	4.38	1.77	0.7
	% Change	5.77	-1.21	23.58	114.69	86.35	7.81	-21.91	89.53	#NUM!
UK	Rank 2024	7	10	2	19	17	3	19	19	6
	Value 2024	130	6.42	7.99	2.26	1.92	7.2	6.36	2.09	6.68
	% Change	10.17	0.21	-1.94	2.2	57.95	-0.99	8.54	21.96	23.26
US	Rank 2024	4.5	4	3	11	39	2	5	16	11
	Value 2024	135	7.64	6.79	3.97	0.82	7.21	7.89	2.4	5.66
	% Change	3.85	0.28	8.85	-3.19	49.5	4.13	0	3.04	8.66

¹ EU27 denotes the EU27 as an aggregate block (excluding intra-EU trade).

Source: Authors' calculation.

Annex 5. Title Definitions of IOI indicators

INDICATOR: PCT_POP		
	Numerator	Denominator
Definition	Number of PCT patent applications by country of residence of the inventor	Millions Population
Source	OECD MSTI (MSTI_PUB/P_PCT.) if available (PATS_IPC/PCT_A.INVENTORS..TOTAL.PRIORITY) OECD REGPAT otherwise	World Bank sp.pop.totl
Notes		
Most recent year used [Nr. Years lag vs. 2024]	2022 [2]	

INDICATOR: TRA_POP		
	Numerator	Denominator
Definition	Number of trademark classes in total trademark resident applications in direct and Madrid system by origin (equivalent count).	Millions of Population
Source	WIPO IP Statistics Data Center: https://www3.wipo.int/ipstats/index.htm?tab=trademark Indicator: "5 – Class count in total applications (direct and via the Madrid system)". Report type: "Total count by applicant's origin (equivalent count)". Type = "Resident".	World Bank sp.pop.totl
Notes	Missing time-series for EL and IE.	
Most recent year used [Nr. Years lag vs. 2024]	2023 [1]	

INDICATOR: KIABI		
	Numerator	Denominator
Definition	Employment in knowledge-intensive business industries	Total employment
Sources	Eurostat, htec_kia_emp2 (DOI: https://doi.org/10.2908/HTEC_KIA_EMP2); ILOSTAT, EMP_TEMP_SEX_EC2_NB_A; Japan Statistical Office, LFS; US BLS CPB; UK BRES; OECD, SSIS_BSC_ISIC4; KOSIS; Central Bureau of Statistics of Israel; New Zealand Statistical office	
Notes	US, JP, UK: data reporting discontinued on Eurostat website; figures were re-computed using national sources, following methodology described by Eurostat htec_esms. KR: Korean Office Statistical Information Service (KOSIS) from 2016-2021 IL: Central Bureau of Statistics of Israel, from 2016-2022 NZ: New Zealand Statistical office, from 2016-2020, figures re-computed following manual matching between ANZIC (Australian and New Zealand Standard Industrial Classification Revision 2.0) and NACE Rev. 2 No data available for CN.	
Most recent year used [Nr. year lag vs 2024]	2024 [0]	

INDICATOR: GOOD_VA		
	Numerator	Denominator
Definition	Domestic value added content of medium-high and high-tech manufacturing exports. Medium-high and high tech industries: ISIC Rev 4 Divisions 20,21,26,27,28,29,30.	Total manufacturing exports Manufacturing sectors: ISIC Rev. 4 Division 10-33.
Sources	OECD TIVA_2022_C1, EXGR_DVASH	
Notes	Data is missing for MK and RS.	
Most recent year used [Nr. year lag vs 2024]	2022 [2]	

INDICATOR: SERV_VA		
	Numerator	Denominator
Definition	Domestic value added content of knowledge-intensive service exports Knowledge intensive service sectors: ISIC Rev. 4 Divisions: 50,51, 58, 59, 60, 61, 62, 63, 64, 65, 66, 69, 70, 71, 72, 73,74,75, 84, 85, 86, 87, 88, 90, 91, 92, 93	Total service exports Service sectors: ISIC Rev. 4 Division 45-95.
Sources	OECD TIVA_2022_C1, EXGR_DVASH	
Notes	Sectors 78 and 80 are not considered as they are aggregated into group 77-82. Data is missing for MK and RS.	
Most recent year used [Nr. year lag vs 2024]	2022 [2]	

INDICATOR: ENT		
	Numerator	Denominator
Definition	Number of innovation-active enterprises Innovation-active firms include product and/or process innovative firms, as well as firms with only innovation activities	Total number of enterprises.
Sources	Eurostat Community Innovation Survey OECD Innovation Indicators database	
Notes	Figures are based on Eurostat CIS core industries (NACE Rev. 2 sections & divisions B-C-D-E-46-H-J-K-71-72-73). Data is missing for CA, IL, and MX.	
Most recent year used [Nr. year lag vs 2024]	2022 [2]	

Annex 6. Contextual indicators

Table 11. Country performance and structure of the economy.

country	GDP per capita (PPS)				FDI inflows (as % of GDP)				Employment in the manufacturing sector, as a percentage of total employment				Employment in the service sector, as a percentage of total employment			
	2024	2023-24 Growth rate (%)	2022-24 Growth rate (%)	2014-24 Growth rate (%)	2024	2023-24 Growth rate (%)	2022-24 Growth rate (%)	2014-24 Growth rate (%)	2024	2023-24 Growth rate (%)	2022-2024 Growth rate (%)	2014-2024 Growth rate (%)	2024	2023-24 Growth rate (%)	2022-24 Growth rate (%)	2014-24 Growth rate (%)
AT	46,880.2	1.5	5.4	35.3	1.8	212.1	-41.2	357.9	15.9	-4.2	-5.4	1.3	72.3	1.5	3.1	4.0
BE	46,523.4	2.8	9.4	43.8	-5.4	1142.9	-367.5	89.4	11.3	-1.7	-0.9	-11.7	79.5	0.0	-0.6	2.7
BG	26,497.8	9.1	18.0	105.5	3.1	-42.7	-44.2	62.8	17.2	-6.0	-5.5	-12.2	67.0	3.6	5.0	6.5
CY	28,027.9	5.8	15.5	82.7	-51.2	-18.6	-1	-121.9	6.1	-1.6	-4.7	-25.6	81.5	0.0	1.0	3.0
CZ	36,479.9	5.1	12.6	55.4	3.8	9.2	24.9	-1.4	24.6	-2.8	-5.4	-8.2	61.8	0.5	1.5	4.4
DE	46,417.5	3.3	8.6	35.1	1.0	-40.3	-51.2	133.4	18.2	-0.5	-4.2	-7.1	72.5	0.4	1.0	2.8
DK	50,186.2	5.5	3.9	46.2	4.3	96.3	-37.2	131.6	11.2	0.0	-0.9	-5.9	78.7	0.3	0.1	0.9
EE	31,768.5	4.2	6.4	50.7	-8.1	-162.4	-265.1	-222.7	15.9	-8.1	-12.6	-12.6	70.4	0.6	2.5	6.8
EL	28,041.9	5.2	15.3	47.9	2.5	29.4	-30.9	117.7	9.8	-1.0	-2.0	8.9	72.6	0.1	-0.8	1.5
ES	36,408.7	5.5	14.9	50.5	2.0	-25.3	-56.3	-14.5	12.0	-0.8	-1.6	-2.4	76.4	-0.1	0.5	0.1
FI	40,766.3	2.0	6.0	36.9	0.8	-831.8	-81.9	-86.8	12.8	-1.5	0.8	-5.2	75.2	1.1	1.1	2.3
FR	39,133.1	3.3	11.2	35.8	1.6	178.3	-59.0	830.2	10.8	-2.7	-0.9	-11.5	76.6	-1.3	-0.5	1.1
HR	31,168.4	3.8	18.5	91.5	5.0	27.3	-8.6	-5.7	15.9	-5.4	-8.6	-6.5	68.4	1.5	3.5	7.2
HU	30,390.4	3.6	10.5	64.4	-15.7	-52.6	1354.0	-271.8	20.5	-1.4	-2.4	-5.1	64.9	0.6	0.9	0.5
IE	88,362.6	6.0	3.6	134.5	-6.7	-73.5	3.9	-117.1	10.6	-1.9	-5.4	-5.4	77.6	0.0	1.2	0.4
IT	38,982.2	3.3	10.6	48.3	0.9	-49.7	-68.8	17.1	18.2	-0.5	-1.6	-2.2	69.9	0.1	0.7	0.6
LT	35,002.7	5.2	11.2	74.4	4.2	-11.8	24.4	472.8	16.1	0.6	1.9	6.6	69.1	0.1	0.4	4.5
LU	96,352.2	2.9	7.2	27.7	113.7	-1173.3	-129.0	290.1	2.4	-25.0	-20.0	-55.6	88.0	2.2	2.1	2.7
LV	28,114.0	3.2	12.3	70.7	3.5	-12.3	8.6	0.8	12.2	-3.2	-5.4	-9.0	71.0	1.1	2.3	3.5
MT	44,263.9	6.2	18.6	75.4	174.8	54.4	38.7	13233.2	9.7	-4.0	-1.0	-24.2	81.1	0.0	-0.6	4.0
NL	53,220.2	6.5	10.1	49.2	-1.4	-95.3	-194.1	-108.7	8.0	-3.6	-4.8	-14.0	82.4	0.1	0.1	9.6
PL	32,274.0	6.4	11.7	76.9	2.3	-44.7	-61.1	-39.3	19.2	2.1	-1.0	0.0	62.8	0.8	2.8	8.5
PT	32,798.7	5.6	17.4	59.1	4.3	8.2	-16.0	-21.5	16.2	-1.8	-4.7	-2.4	72.4	0.6	0.4	5.8
RO	31,086.0	5.6	17.9	110.2	1.9	-22.6	-50.4	-0.9	18.8	-3.6	-5.1	1.6	56.1	2.0	0.2	31.4
SE	44,927.0	4.1	8.9	33.8	4.7	50.2	-49.2	-552.4	9.5	-3.1	1.1	-11.2	81.0	0.2	0.6	2.5
SI	36,336.9	3.7	13.5	65.8	2.7	28.6	-26.0	30.8	23.0	1.8	5.0	1.3	64.1	0.6	-1.8	7.7
SK	29,945.8	5.3	16.3	43.5	2.5	-1132.0	-39.4	-808.0	21.9	-1.4	-8.4	-6.0	62.8	0.2	1.9	3.0
EU	40,019.3	4.2	10.8	49.4	1.5	-208.7	8.2	-48.9	15.3	-1.3	-3.2	-6.7	72.1	0.1	0.8	4.0

¹ FDI inflows in CY as % of GDP in 2022 were very close to zero, preventing the estimation of a valid growth rate.

Source: Eurostat and World Bank's World Development Indicators, accessed in December 2025.

Table 12. Orientation of the economy toward knowledge-intensive sectors.

country	Employment in the high and medium-high tech manufacturing sectors (% of total manufacturing employment)				Employment in the knowledge-intensive service sector (% of total service employment)				The share of medium- and high-tech products in total product exports (GOOD)				Knowledge-intensive services exports as share of total service exports (SERV)			
	2024	2023-24 Growth rate (%)	2022-24 Growth rate (%)	2014-24 Growth rate (%)	2024	2023-24 Growth rate (%)	2022-24 Growth rate (%)	2014-24 Growth rate (%)	2024	2023-24 Growth rate (%)	2022-24 Growth rate (%)	2014-24 Growth rate (%)	2024	2023-24 Growth rate (%)	2022-24 Growth rate (%)	2014-24 Growth rate (%)
AT	6.5	1.6	1.6	10.2	41.3	1.7	3.8	9.0	58.2	3.5	5.8	2.0	47.8	1.8	-0.6	7.3
BE	4.1	-2.4	-2.4	-14.6	50.1	0.0	-0.4	5.5	51.7	2.8	9.6	8.5	73.4	2.0	2.4	9.4
BG	4.0	-13.0	-11.1	8.1	34.0	5.3	7.3	10.7	35.4	-2.5	4.8	23.3	57.6	4.7	6.1	42.2
CY	0.8	-11.1	-20.0	-11.1	42.6	-1.2	0.9	6.8	21.0	0.0	-34.4	-68.0	85.2	0.8	-1.5	18.4
CZ	10.5	-1.9	-4.5	-6.2	36.7	1.1	3.1	12.6	69.2	1.3	5.7	8.3	50.4	-0.3	-0.4	18.2
DE	9.6	1.1	2.1	-1.0	43.7	1.2	2.3	10.1	67.1	-0.7	3.5	1.6	74.1	-0.5	-1.8	3.2
DK	5.4	1.9	8.0	5.9	49.8	0.0	1.4	-0.4	52.0	1.3	5.5	12.3	78.3	1.8	-7.1	0.3
EE	3.8	-7.3	-9.5	8.6	39.8	0.3	3.4	10.2	41.7	2.4	9.4	2.3	67.5	-0.1	2.6	48.8
EL	1.4	-12.5	-17.6	16.7	37.7	-0.5	-1.3	4.1	24.3	3.6	14.2	26.5	53.1	-0.6	-10.0	3.3
ES	4.0	0.0	0.0	2.6	38.1	0.0	0.0	5.2	44.8	-1.3	2.9	-0.6	42.0	-7.6	-9.1	10.5
FI	5.2	4.0	13.0	4.0	47.4	1.7	3.3	5.1	45.8	0.1	8.5	16.0	78.5	5.2	0.8	4.2
FR	3.7	-5.1	-2.6	-15.9	46.6	-1.9	0.4	2.0	54.5	0.0	4.0	-4.7	66.4	1.8	-2.4	2.6
HR	3.3	-2.9	-8.3	0	38.1	2.4	7.3	12.7	38.4	0.0	18.6	9.4	24.3	4.4	8.0	19.7
HU	9.1	2.2	0.0	2.2	37.2	2.5	1.6	4.2	70.7	-0.2	9.4	5.6	52.3	-0.5	-5.6	8.7
IE	4.0	-4.8	-4.8	-20.0	49.6	2.1	3.5	8.8	68.0	6.9	10.6	39.7	92.0	1.7	1.6	7.6
IT	6.1	-1.6	-1.6	1.7	35.5	-0.3	1.7	2.9	51.3	-0.5	4.2	0.1	50.6	-0.5	-1.9	-1.3
LT	2.6	-3.7	-7.1	36.8	40.0	3.9	5.0	22.0	36.2	-0.9	4.0	5.7	40.3	1.2	20.1	91.9
LU	0.6	20.0	-14.3	-53.8	60.2	3.6	2.7	3.3	41.7	-0.2	1.9	-13.4	87.5	3.5	2.3	-1.0
LV	2.0	5.3	-4.8	25.0	40.3	5.2	6.1	12.9	32.1	-3.0	7.8	4.0	59.1	-1.0	-0.5	24.9
MT	2.9	3.6	16.0	-27.5	49.6	0.2	-1.0	11.7	62.5	0.0	5.8	0.0	44.3	-4.3	26.5	-23.5
NL	2.8	-3.4	-3.4	3.7	53.3	0.6	0.4	14.4	50.6	0.9	8.8	18.3	76.7	0.0	0.0	-0.6
PL	5.8	9.4	11.5	11.5	36.0	1.7	6.5	14.6	49.2	-1.0	3.3	0.5	52.6	4.6	11.3	35.2
PT	3.8	0.0	0.0	26.7	39.3	1.6	-0.3	12.6	41.4	-1.2	4.3	15.4	41.9	1.4	1.1	-3.5
RO	6.9	-1.4	1.5	30.2	25.2	3.3	-1.9	26.0	58.1	6.5	12.4	15.5	54.8	2.7	3.6	23.4
SE	4.5	-4.3	0.0	4.7	57.1	0.0	0.4	8.8	56.4	0.7	9.4	7.9	82.0	3.0	5.2	1.9
SI	9.0	4.7	2.3	4.7	37.6	1.1	-7.8	8.4	71.2	6.0	13.7	28.4	40.8	4.0	0.4	17.1
SK	10.1	-1.9	-10.6	7.4	36.6	0.8	-0.5	7.6	68.9	-1.5	4.5	6.0	47.2	4.3	13.2	33.7
EU	5.9	0.0	0.0	-1.7	41.5	0.5	1.7	8.4	62.9	1.0	5.4	5.1	77.0	1.7	-0.8	5.5

Source: Eurostat, accessed in December 2025.

Table 13. Entrepreneurship.

country	Birth rate (number of enterprise births on the total number of active enterprises)			Employment share of enterprise births			Employment in fast-growing enterprises, as a percentage of total employment (DYN_rev)		
	2023	2022-23 Growth rate (%)	2021-23 Growth rate (%)	2020	2023	2022-23 Growth rate (%)	2023	2022-23 Growth rate (%)	2021-23 Growth rate (%)
AT	6.2	0.2	3.8	1.3	0.0	4.9	9.0	14.9	23.5
BE	8.2	-9.0	5.9	2.4	-12.9	4.7	5.4	11.7	27.9
BG	10.7	7.1	6.9	2.5	-8.2	-6.1	15.4	11.8	14.5
CY	9.7	-1.6	-4.7	2.0	-2.0	-20.2	6.4	92.7	128.0
CZ	9.1	0.6	3.4	2.4	2.6	8.3	11.3	0.5	4.4
DE	8.4	5.6	7.9	1.1	-0.9	-1.7	12.9	-1.8	-1.1
DK	7.3	-2.2	-35.5	1.2	1.7	-19.0	22.1	39.9	25.4
EE	14.3	-14.3	-43.9	5.1	-12.3	-35.1	16.8	1.6	15.4
EL	8.5	-14.0	17.0	4.6	-32.8	48.7	28.4	52.5	30.3
ES	9.1	5.4	-0.9	2.4	-3.6	-17.9	11.6	-23.1	-26.8
FI	9.0	1.2	0.9	0.9	-5.4	-15.5	25.3	63.4	53.1
FR	14.0	-1.7	-13.2	2.0	-2.9	-21.3	6.3	37.1	40.1
HR	12.5	8.5	13.6	3.0	6.0	9.1	13.8	18.6	26.7
HU	9.4	0.2	-21.2	2.9	3.5	-24.4	19.0	16.8	25.7
IE	11.9	-2.9	16.3	3.1	-8.7	57.0	30.0	41.4	45.5
IT	7.8	-1.5	-3.6	2.3	7.8	5.9	12.4	40.5	43.0
LT	19.6	7.4	1.6	3.6	5.8	8.0	16.7	12.8	-3.4
LU	9.0	-15.1	4.4	1.1	-10.6	5.8	15.3	26.8	33.3
LV	12.9	-9.9	-1.2	2.8	-8.8	-20.0	12.0	-24.2	-9.6
MT	17.1	19.7	18.6	4.5	13.7	6.7	24.5	22.4	10.5
NL	9.4	1.4	-14.5	1.5	1.3	-6.8	13.8	-8.7	-2.8
PL	11.5	-6.2	-6.6	3.8	-9.1	0.3	14.3	0.1	-1.4
PT	16.8	0.5	17.0	5.9	2.1	18.3	19.3	19.9	22.9
RO	12.4	2.7	7.9	3.9	16.9	8.1	9.1	-37.0	-61.9
SE	8.2	-9.9	0.4	1.5	4.9	-37.5	20.9	-10.5	-5.7
SI	10.5	-6.6	-3.0	2.8	-6.0	0.4	10.0	15.7	16.1
SK	11.7	-2.6	2.7	4.5	-2.4	9.5	13.0	10.2	33.9
EU	10.5	-0.6	-3.1	2.3	-2.6	-3.0	12.9	5.4	3.8

Source: Eurostat, accessed in December 2025.

Table 14. Innovation.

country	R&D investment (PPS, 2005 constant prices)				R&D expenditure, as a percentage of GDP				Employment in innovation-active firms, % of total employment			Share of turnover from new-to-market products ¹		
	2024	2023-24 Growth rate (%)	2022-24 Growth rate (%)	2014-24 Growth rate (%)	2024	2023-24 Growth rate (%)	2022-24 Growth rate (%)	2014-24 Growth rate (%)	2022	2020-22 Growth rate	2014-22 Growth rate	2022	2020-22 Growth rate	2014-22 Growth rate
AT	9,571.1	0.6	1.6	18.1	3.3	1.2	2.8	4.8	77.0	-4.6	-4.1	5.0	-12.3	-40.5
BE	12,638.7	5.0	7.8	67.7	3.4	3.7	4.7	42.4	84.1	-2.2	4.7	5.7	3.6	3.6
BG	807.5	0.1	7.4	29.2	0.8	-2.5	2.7	-2.5	51.8	-13.5	-3.7	2.7	-3.6	-72.2
CY	182.2	2.1	-0.2	108.3	0.7	-1.5	-7.1	27.5	57.8	-24.4	-6.6	6.9	40.8	43.8
CZ	4,860.6	1.2	-2.4	16.0	1.8	0.0	-3.7	-6.2	67.4	-11.7	-1.7	5.7	-6.6	-61.5
DE	84,609.4	-0.4	1.5	21.1	3.1	0.0	3.0	11.0	85.5	-1.0	-0.9	4.3	26.5	-8.5
DK	5,901.0	1.7	8.9	27.0	3.0	-2.0	4.5	3.1	66.0	-3.2	-3.5	9.8	108.5	-5.8
EE	507.0	8.3	9.1	69.7	2.0	8.7	12.4	41.1	68.6	-10.9	51.8	5.5	0.0	-80.1
EL	3,232.5	5.2	8.5	109.4	1.5	2.7	4.1	81.2	76.9	-6.8	7.0	7.5	-22.7	-26.5
ES	18,390.9	3.6	12.9	49.4	1.5	0.7	7.1	22.0	64.8	11.3	3.2	8.4	31.3	-39.6
FI	5,058.5	4.6	7.5	11.6	3.2	4.2	8.1	1.9	76.6	-6.4	0.3	6.3	26.0	-1.6
FR	42,265.8	1.3	0.7	10.2	2.2	0.0	-1.8	-1.8	79.2	-3.4	-1.1	3.4	-5.6	-66.3
HR	1,023.8	2.6	2.5	143.7	1.4	-1.5	-4.9	75.3	62.7	-13.3	6.3	2.3	-41.0	-66.2
HU	2,567.6	-4.8	-6.4	27.3	1.3	-5.1	-5.8	-2.2	53.7	0.4	2.1	6.6	153.8	-48.8
IE	4,606.6	-8.4	-10.1	95.9	1.4	-10.4	-9.8	-6.8	71.1	-6.4	-6.4	8.3	-54.9	-66.7
IT	20,778.6	1.5	2.9	15.5	1.4	0.7	0.7	3.8	78.7	9.8	10.1	5.2	20.9	-49.0
LT	710.9	3.3	2.4	38.6	1.0	0.0	-1.9	1.0	77.7	2.0	14.9	3.0	-30.2	-67.4
LU	389.5	-5.7	-6.4	-2.1	1.0	-6.6	-7.5	-18.9	62.9	-3.5	-19.9	2.2	57.1	-86.7
LV	305.1	12.6	12.5	55.6	0.9	12.2	13.6	29.6	52.0	0.8	4.8	5.8	93.3	-52.8
MT	100.2	-6.7	2.3	42.4	0.5	-12.1	-12.1	-25.0	56.8	-4.5	-10.1	6.9	122.6	13.1
NL	15,371.0	0.3	5.7	28.9	2.3	-0.4	5.0	6.5	70.4	-0.3	-8.3	6.8	38.8	-55.6
PL	12,309.7	-6.7	1.0	113.4	1.4	-9.6	-2.1	48.4	63.8	4.8	22.2	2.4	-33.3	-67.1
PT	4,031.2	4.7	7.2	65.1	1.7	2.4	1.8	33.3	70.2	-0.3	1.0	5.7	32.6	11.8
RO	1,355.1	-11.0	2.1	65.5	0.5	-11.5	0.0	21.1	19.3	-9.8	-16.8	3.7	270.0	-80.4
SE	12,392.1	-1.2	2.5	36.1	3.6	-1.9	2.0	14.4	76.5	-5.7	3.8	9.9	57.1	45.6
SI	1,221.1	3.3	7.2	20.7	2.2	1.4	2.9	-9.6	75.7	0.0	5.6	3.7	2.8	-61.1
SK	1,252.5	-3.4	4.6	42.0	1.0	-5.8	0.0	12.6	57.1	-7.5	7.9	12.1	1.7	-62.2
EU	266,440.1	0.3	2.9	28.1	2.2	-0.9	0.9	7.2	72.9	-0.4	2.2	5.2	13.0	-40.9

¹ Data on the share of new-to-the market products refer to 2018 for Estonia.

Source: Eurostat, accessed in December 2025.

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